



MANAGERIAL BUSINESS ANALYTICS

Self-Instructional Manual

The Self-Instructional Manual (SIM) for Managerial Business Analytics (MBA6653) is designed to develop students' competencies in applying statistical analysis and data analytics to support evidence-based managerial decision-making. The course integrates descriptive, diagnostic, predictive, and prescriptive analytics within the Malaysian and ASEAN business context.

About This Manual

This manual comprises six learning units covering the full analytics lifecycle — from data collection and exploration to diagnostic, predictive, and prescriptive analytics, culminating in data visualisation and storytelling. Each unit includes learning outcomes, core concepts, worked examples, Malaysian and Sarawak-contextualised case studies, self-reflection prompts, self-check questions, and discussion questions aligned with the course learning outcomes. The SIM promotes independent learning and critical engagement with analytics in organisational contexts.



MBA6653 | Major (Core) | Credit 3 | 120 SLT | Semester 1

Lecturers: Dr. Colin Wong Koh King; Faizul Bin Ayupp Dino; AP Michael Tinggi

INSPIRING MINDS, SHAPING FUTURE

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PREFACE

MANAGERIAL BUSINESS ANALYTICS

(MBA6653)

This Self-Instructional Manual (SIM) for Managerial Business Analytics (MBA6653) has been developed to equip postgraduate students with the analytical competencies required to navigate the increasingly data-driven landscape of modern management. In an era where organisations across Malaysia and the ASEAN region are rapidly embracing digital transformation, the ability to collect, analyse, interpret, and communicate data insights has become a critical managerial skill.

The course is structured around six learning units that collectively address the full spectrum of business analytics: from foundational data literacy and exploratory analysis, through diagnostic and predictive modelling, to prescriptive decision-support and data storytelling. Each unit is carefully sequenced to build upon prior knowledge, ensuring that learners develop both conceptual understanding and practical capability.

A distinguishing feature of this SIM is its strong Malaysian and Sarawak contextualisation. Case studies and examples draw from organisations such as Sarawak Energy Berhad, PETROS, Sarawak Digital Economy Corporation (SDEC), the Sarawak tourism sector, Malaysian e-commerce platforms, and i-CATS University College itself. These locally grounded examples make abstract analytics concepts tangible and directly applicable to the learner's professional environment.

The SIM is designed for self-paced Open and Distance Learning (ODL). Each unit includes self-check questions, self-reflection prompts, discussion questions aligned with course learning outcomes, and a summary section to reinforce learning. Learners are encouraged to engage actively with the exercises and to draw connections between analytics theory and the real-world business challenges they encounter in their careers.

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Course Information

Course Code	MBA6653
Course Title	Managerial Business Analytics
Credit Value	3
Total Student Learning Time (SLT)	120 hours
Classification	Major Core
Semester	2
Pre-requisite	None

Course Synopsis

This course covers the use of statistical analysis and business analytics for managerial decision-making across a wide range of functional areas. Students learn how to gather, prepare, and analyse data; interpret results; and communicate insights effectively to support evidence-based decisions. The course emphasises descriptive, diagnostic, predictive, and prescriptive analytics, with applications drawn from Malaysian and ASEAN organisations. Students will also develop competencies in data visualisation and storytelling to present analytical findings to managerial audiences effectively.

Course Learning Outcomes (CLO)

- CLO1: Analyse complex datasets using appropriate analytical tools and techniques to provide meaningful insights for more effective and evidence-based business decisions. (C6, PLO1)
- CLO2: Explain information about data, analysis, and insights from the data clearly and persuasively, in both written and oral form. (A4, PLO3)
- CLO3: Evaluate the potential for data-driven analytics and evidence-based practice to solve business problems and enhance competitive advantage. (A6, PLO7)

Assessment Structure

Component	Type	Weightage	CLO Alignment
CA1	Project	35%	CLO1, CLO2
CA2	Case Study	35%	CLO1, CLO3
Final Assessment	Individual Assignment	30%	CLO1, CLO2, CLO3

How to Use This Manual

This Self-Instructional Manual is designed for self-paced, independent study as part of the Open and Distance Learning (ODL) programme at i-CATS University College. Each learning unit is structured to guide you progressively through core analytics concepts, practical applications, and reflective activities.

Recommended approach for each learning unit:

1. Read the Learning Outcomes at the start of each unit so you know what you are expected to achieve by the end.
2. Work through the content sections systematically — do not skip ahead, as each section builds on prior knowledge.
3. Engage actively with the Key Concept and Managerial Insight boxes — these highlight the most important ideas for managerial application.
4. Attempt the Worked Examples on your own before reading the solutions — this develops analytical reasoning.
5. Complete the Pause and Reflect prompts honestly — these are designed to connect theory to your own organisational experience.
6. Test yourself using the Self-Check Questions at the end of each unit before reviewing the Answer Key.
7. Use the Discussion Questions to prepare for tutorials and online discussions.
8. Review the Summary after completing each unit to consolidate your learning.

The glossary at the back of this manual provides definitions of all key analytics terms used throughout the SIM. Refer to it whenever you encounter an unfamiliar term. The full reference list at the back provides sources for further reading and self-directed exploration.

Course Schedule

The following schedule maps the six learning units across 14 weeks of study, totalling 120 Student Learning Time (SLT) hours. This course is delivered in online independent learning mode. Students are expected to manage their own study time according to this schedule and to engage with all activities, assessments, and resources indicated.

Week	LU	Topics Covered	SLT (hrs)
1	LU1	Data for Business — Introduction, Sampling, Data Types	8
2	LU1	Data for Business — Stages of Analytics, CRISP-DM, Business Insights	8
3	LU2	Exploratory Data Analytics — Descriptive Measures, Central Tendency	8
4	LU2	Exploratory Data Analytics — Dashboards, Scorecards, EDA Applications	8
5	LU3	Diagnostic Analytics — Dispersion, Hypothesis Testing	8
6	LU3	Diagnostic Analytics — ANOVA, t-test, Correlation, Trend Analysis	8
7	LU4	Predictive Analytics — Classification, Regression	8
8	LU4	Predictive Analytics — Clustering, Association Rules, Applications	8
9	LU5	Prescriptive Analytics — Machine Learning Overview, Simulations	8
10	LU5	Prescriptive Analytics — Logic-Based Models, Optimisation, Applications	8
11	LU6	Data Visualisation — Principles, Chart Selection, Design	8

12	LU6	Data Visualisation — Dashboards, Storytelling with Data	8
13	Revision	Review of all LUs; CA2 Case Study preparation	6
14	Assessment	Final Assessment preparation; submission of CA2	6

Note: Total SLT = 120 hours (112 hours of guided and independent study + 8 hours of assessment). CA1 Project is due at the end of Week 6. CA2 Case Study is due at the end of Week 14. The Final Assessment is a take-home individual assignment submitted within two weeks of the final teaching week.

TOPIC

1

DATA FOR BUSINESS

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LEARNING OUTCOMES

By the end of this topic you should be able to:

- **Explain the role and types of data in supporting managerial decision-making**
- **Describe the stages of the data analytics lifecycle and the CRISP-DM framework**
- **Distinguish between types of variables and appropriate data collection methods**
- **Apply foundational analytics concepts to real-world Malaysian business scenarios**



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LO Learning Objectives

By the end of this topic you should be able to:

- **Explain the fundamental stages of data analytics** and their application in business decision-making.
- **Apply appropriate sampling and data collection techniques** to gather business-relevant datasets.
- **Distinguish between types of variables** and select suitable data types for analytical purposes.
- **Describe the CRISP-DM framework** and apply it to structure a business analytics project.

1 Introduction to Business Analytics

Business analytics is the practice of iterative, methodical exploration of an organisation's data, with emphasis on statistical analysis. It is used to drive fact-based management decision-making. Analytics moves management away from intuition and gut-feel toward evidence-based decisions, enabling organisations to respond faster and more accurately to market changes, customer behaviour, and operational challenges.

There are three broad categories of analytics users in the business world. Analytics-at-scale organisations — such as Amazon and Grab — deploy analytics as a core competitive weapon, embedding data-driven decision-making into every function from supply chain to customer engagement. Analytics-as-support organisations — which include most Malaysian listed companies such as Maybank, Petronas, and Telekom Malaysia — use analytics to support specific business functions, particularly finance, marketing, and operations. Analytics-emergent organisations — including small Malaysian SMEs and Sarawak rural businesses — are just beginning their analytics journey, often relying on basic spreadsheet tools and moving toward more structured data practices.

SUMMARY

Managerial Insight

In 2023, Malaysian companies in the top quartile of analytics maturity reported 23% higher return on equity compared to industry peers, according to Deloitte's Southeast Asia Analytics Survey. This underlines the competitive necessity of analytics capability for Malaysian organisations at all stages of their growth journey.

2 Types of Variables and Data

Data can be broadly classified into quantitative data and qualitative data. Quantitative data is numerical and can be measured on a scale — for example, monthly sales revenue, number of customer transactions, or employee headcount. Qualitative data (also called categorical data) describes attributes or categories without inherent numerical value — for example, customer segment type, product colour, or geographical region.

Within quantitative data, a further distinction is made between continuous variables and discrete variables. Continuous variables can take any value within a range, including fractional values — such as temperature, weight, or time elapsed. Discrete variables can only take specific integer or countable values — such as the number of items sold, number of complaints received, or headcount. This distinction matters because different statistical techniques are appropriate for each variable type.

Scales of Measurement

Variables can be measured at four levels of measurement, each with different analytical capabilities. Understanding the scale of measurement is critical for selecting the appropriate statistical technique.

Scale	Description	Example	Mathematical Operation
Nominal	Categories with no order	Customer segment (Gold/Silver/Bronze)	Count, mode
Ordinal	Categories with rank order	Customer satisfaction rating (1–5)	Count, mode, median
Interval	Equal intervals, no true zero	Temperature in Celsius	Mean, standard deviation
Ratio	Equal intervals, true zero	Monthly sales revenue (RM)	All operations including ratio

An additional major classification of data is structured versus unstructured data. Structured data is organised in a predefined format — typically rows and columns in relational databases or spreadsheets — and is directly amenable to standard statistical analysis. Unstructured data lacks a predefined format and includes text documents, images, video, audio recordings, and social media posts. For example, Shopee Malaysia generates millions of unstructured product reviews daily; analysing this text data requires natural language processing (NLP) techniques rather than traditional statistical methods. The majority of enterprise data today — estimated at over 80% — is unstructured.

Key Characteristics of Good Data Quality:

- Accuracy — data correctly reflects the real-world entity or event it represents
- Completeness — all required data values are present, with minimal missing entries
- Consistency — data is uniform across systems and time periods, with no contradictions
- Timeliness — data is current and available when needed for decision-making
- Validity — data conforms to defined formats, ranges, and business rules

Data used in analytics can be either primary or secondary in origin. Primary data is collected directly by the researcher or organisation for the specific purpose of the analysis — through surveys, interviews, observations, or experiments. Primary data is tailored to the research question but is expensive and time-consuming to collect. Secondary data is data that was originally collected for another purpose and is being reused — such as government statistics, industry reports, financial databases, or internal CRM records. Secondary data is faster and cheaper to obtain but may not perfectly match the analytical requirements.

Sampling Methods

Since it is rarely feasible to collect data from an entire population, analysts typically work with samples. The choice of sampling method affects the reliability and validity of conclusions drawn from the sample.

Sampling Method	Description	When to Use	Example
Simple Random	Every member of the population has an equal chance of selection	Homogeneous populations; when a complete list is available	Randomly selecting 100 customers from a database of 10,000
Stratified	Population divided into subgroups (strata); random sample taken from each	Heterogeneous populations; when subgroup representation is important	Sampling separately from Gold, Silver, and Bronze customer tiers
Systematic	Every nth element selected from a list after a random start	When a complete ordered list is available	Selecting every 50th invoice from a monthly transaction log

Sampling Method	Description	When to Use	Example
Cluster	Population divided into clusters; entire clusters randomly selected	Geographically dispersed populations; cost-efficiency required	Selecting five Sarawak districts and surveying all businesses within them
Convenience	Sample selected based on ease of access	Exploratory research; preliminary pilot studies	Surveying walk-in customers at a Kuching retail outlet

Sampling error is the natural variation between a sample statistic and the true population parameter — it arises from the inherent randomness of the sampling process and can be reduced by increasing sample size. Non-sampling error, in contrast, arises from systematic issues such as measurement error, non-response bias, data entry mistakes, or poorly designed survey questions — these errors do not diminish with larger samples and must be addressed through rigorous data collection protocols.

Data Collection Challenges in the Sarawak Context

Collecting reliable data in Sarawak presents distinctive challenges that analysts must account for in their research design. Geographic dispersion is a primary challenge — Sarawak is the largest state in Malaysia, covering 124,450 square kilometres with many communities accessible only by river or air. Rural internet connectivity remains limited in interior divisions such as Kapit, Belaga, and Ba'Kelalan, making digital survey methods unreliable for these populations. Sarawak is also a multilingual society; survey instruments and interview protocols must account for respondents whose primary languages may be Iban, Malay, Mandarin, Bidayuh, or other indigenous languages, requiring translation and localisation that can introduce additional measurement error.

Self-Reflection



1. Think about a decision your organisation recently made. What data was used to support that decision? Was the data collected systematically, or was it based on ad hoc observations? What additional data might have improved the decision quality?

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Stages of Data Analytics

Business analytics has evolved through five progressive stages, each building on the capabilities of the previous and answering increasingly sophisticated managerial questions. Understanding these stages helps managers allocate analytics investment appropriately and identify where their organisation currently sits on the analytics maturity spectrum.

Stage 1: Descriptive Analytics — "What happened?"

Descriptive analytics involves summarising and aggregating historical data to understand what has occurred in the past. It is the most widely used form of analytics and forms the foundation for all higher-level analysis. Techniques include frequency tables, charts, dashboards, and summary statistics. A practical Malaysian example is Maybank's use of monthly transaction dashboards that aggregate millions of daily transactions into branch-level and product-level performance summaries, enabling management to monitor performance against targets.

Stage 2: Diagnostic Analytics — "Why did it happen?"

Diagnostic analytics applies statistical techniques — such as hypothesis testing, ANOVA, correlation, and root cause analysis — to identify the factors that caused observed outcomes. It transforms the "what" of descriptive analytics into the "why" of managerial understanding. AirAsia, for example, uses diagnostic analytics to identify root causes of on-time performance drops, examining correlations between ground handling delays, aircraft rotation schedules, weather disruptions, and crew availability to pinpoint the primary drivers of performance variance.

Stage 3: Predictive Analytics — "What will happen?"

Predictive analytics uses statistical models and machine learning algorithms to forecast future outcomes based on historical patterns. Common techniques include regression analysis, time-series forecasting, decision trees, and neural networks. Sarawak Energy Berhad (SEB) applies predictive analytics to forecast peak electricity demand across the Sarawak grid, integrating historical load data, weather forecasts, and industrial production schedules to optimise generation planning and minimise costly emergency capacity procurement.

Stage 4: Prescriptive Analytics — "What should we do?"

Prescriptive analytics moves beyond prediction to recommend specific actions that optimise a defined objective. It uses techniques such as linear programming, simulation, and reinforcement learning. Grab Malaysia's driver-rider matching algorithm is a real-world example of prescriptive analytics at scale — the algorithm continuously optimises driver dispatch decisions across thousands of concurrent ride requests, minimising wait times, maximising driver utilisation, and balancing demand across geographic zones in real time.

Stage 5: Cognitive/AI Analytics — "What can we automate?"

Cognitive analytics integrates artificial intelligence, natural language processing (NLP), computer vision, and machine learning to automate complex analytical tasks that previously required human judgment. Malaysian banks — including CIMB, Maybank, and RHB — have deployed NLP-based systems to process loan applications, extracting and interpreting unstructured text from financial documents, correspondence, and supporting evidence to generate credit risk assessments with minimal human intervention.

Stage	Question Answered	Technique Examples	Difficulty Level
Descriptive	"What happened?"	Dashboards, frequency tables, summary statistics	Basic
Diagnostic	"Why did it happen?"	Hypothesis testing, ANOVA, correlation, root cause analysis	Intermediate

Stage	Question Answered	Technique Examples	Difficulty Level
Predictive	"What will happen?"	Regression, time-series forecasting, decision trees	Advanced
Prescriptive	"What should we do?"	Optimisation, simulation, reinforcement learning	Expert
Cognitive/AI	"What can we automate?"	NLP, computer vision, deep learning, generative AI	Expert+

5 Business Analytics Insights

The Data–Information–Insight Hierarchy

Raw data consists of individual facts, measurements, or observations without context — a list of transaction amounts, timestamps, or GPS coordinates. When data is processed, organised, and given context, it becomes information — for example, "total sales by product category for Q3 2024." An insight is a non-obvious, actionable finding derived from information that has strategic or operational significance — for example, "customers who purchase Product A in their first transaction have a 73% probability of making a repeat purchase within 30 days, but this behaviour is concentrated entirely among customers aged 25–34 in urban areas." Insights drive decisions; raw data alone does not.

Characteristics of Valuable Analytics Insights:

- Actionable — the insight suggests a specific management response or decision
- Accurate — derived from clean, validated data using appropriate methodology
- Timely — available when managers need it to make decisions
- Relevant — directly related to a strategic or operational priority
- Understandable — communicable to non-technical managers in plain language

Business Value Drivers of Analytics:

- Cost reduction — identifying inefficiencies, waste, and unnecessary expenditure
- Revenue growth — uncovering cross-sell, up-sell, and new market opportunities
- Risk management — detecting fraud, credit risk, and operational failures earlier
- Customer experience — personalising service delivery and predicting churn
- Operational efficiency — optimising processes, scheduling, and resource allocation

Case Study: Sarawak Energy Berhad (SEB) Digital Analytics

Sarawak Energy Berhad (SEB) has implemented smart metering across its 600,000+ customer base in Sarawak. By analysing smart meter data using descriptive and diagnostic analytics, SEB identified that industrial customers account for 68% of peak demand events across the Sarawak grid. This insight led to the design of a targeted demand response programme that directly engaged the top 200 industrial consumers with variable pricing incentives and load-shifting agreements. The programme reduced peak demand by 12% in 2023, avoiding the need for an estimated RM200 million in new generation capacity investment.

The Sarawak Digital Economy Corporation (SDEC) collaboration with SEB under the Sarawak Digital Economy Blueprint 2030 has further expanded SEB's analytics capability to include predictive load forecasting. This model integrates historical consumption data, real-time weather data from the Sarawak Meteorological Department, and leading economic indicators to generate 72-hour demand forecasts, enabling more precise dispatch planning and fuel cost optimisation. The SEB case illustrates how analytics insights — even at the descriptive and diagnostic level — can generate quantifiable business value in a regulated utility environment.

6

CRISP-DM Framework

CRISP-DM — the Cross-Industry Standard Process for Data Mining — is the most widely adopted methodology for structuring analytics and data mining projects. First published in 1999 and continuously updated, CRISP-DM provides a vendor-neutral, technology-agnostic framework that guides analysts through a structured, iterative project lifecycle. It is applicable across all industries and analytics techniques, from simple regression models to complex deep learning systems.

Phase 1: Business Understanding

The first and most critical phase involves defining the business objectives of the analytics project, establishing success criteria, and translating these into a data mining problem statement. Typical tasks include interviewing stakeholders, reviewing existing reports and strategies, assessing available resources, and developing a project plan. A failure to invest adequately in business understanding is the most common reason analytics projects fail to deliver value.

Phase 2: Data Understanding

This phase involves collecting initial data, exploring its content and structure, and assessing its quality and suitability for the intended analysis. Analysts examine variable distributions, identify missing values and outliers, and develop initial hypotheses about relationships in the data. The outputs of this phase directly inform the scope and feasibility of the modelling approach.

Phase 3: Data Preparation

Data preparation is typically the most time-intensive phase, consuming 60–80% of total project effort. It encompasses data cleaning (handling missing values, correcting errors), data transformation (normalisation, encoding categorical variables), feature engineering (creating new derived variables), and assembling the final analytical dataset. The quality of this phase largely determines the quality of subsequent models.

Phase 4: Modelling

The modelling phase involves selecting appropriate modelling techniques, calibrating model parameters, and building the analytical model. Multiple techniques are often tested and compared. Modelling in CRISP-DM frequently reveals data requirements that necessitate returning to the data preparation phase — reinforcing the iterative nature of the framework.

Phase 5: Evaluation

The evaluation phase assesses whether the model meets the business objectives defined in Phase 1. This goes beyond technical model performance metrics (accuracy, R^2 , AUC) to assess whether the model's outputs are interpretable, actionable, and aligned with business strategy. A thorough review process is conducted to identify any shortcomings before deployment.

Phase 6: Deployment

Deployment involves integrating the model into organisational processes and systems so that its outputs can be consumed by decision-makers. This may range from generating a simple report to embedding a real-time scoring model into an operational system. Documentation, monitoring plans, and maintenance procedures are established to ensure the model continues to perform over time.

Phase	Key Questions	Key Outputs
1. Business Understanding	What is the business problem? What does success look like? What are the constraints?	Project charter, business objective statement, success criteria, project plan
2. Data Understanding	What data is available? Is it sufficient? What quality issues exist?	Data inventory, exploratory analysis report, data quality assessment
3. Data Preparation	How should data be cleaned and structured? What features should be engineered?	Clean analytical dataset, feature definitions, transformation documentation
4. Modelling	Which modelling technique is most appropriate? What are the optimal parameters?	Fitted model(s), model comparison results, parameter settings
5. Evaluation	Does the model meet business objectives? Are there gaps or risks?	Model evaluation report, revised project plan or go/no-go decision

Phase	Key Questions	Key Outputs
6. Deployment	How will the model be used? How will performance be monitored?	Deployed model, monitoring plan, user documentation, maintenance schedule

It is important to understand that CRISP-DM is explicitly iterative — it is not a linear, one-pass process. The evaluation phase frequently reveals that the model does not fully meet business objectives, triggering a return to data preparation or even business understanding. Real-world analytics projects rarely proceed phase by phase without revisiting earlier work; CRISP-DM accommodates and encourages this natural exploratory cycle.

Self-Reflection



2. Select a business problem in your organisation that could benefit from analytics. Apply the CRISP-DM framework: What is the business objective? What data is available? What type of model would be most appropriate? What would "deployment" look like in your organisation?

SUMMARY

Business analytics transforms data into actionable insights to support evidence-based managerial decisions, moving organisations away from intuition toward fact-based management.

Data types include quantitative (continuous and discrete) and qualitative; measurement scales are nominal, ordinal, interval, and ratio — each with different analytical capabilities.

Sampling methods — random, stratified, systematic, cluster, and convenience — each suit different population characteristics and research contexts.

The five stages of analytics — descriptive, diagnostic, predictive, prescriptive, and cognitive — represent a maturity spectrum from basic reporting to AI-driven automation.

CRISP-DM provides a structured, iterative six-phase methodology for analytics projects, widely adopted across industries and applicable to all levels of analytics complexity.

Malaysian and Sarawak organisations — from Maybank and AirAsia to Sarawak Energy Berhad — are increasingly leveraging analytics for competitive advantage and operational excellence.

Multiple Choice Questions (MCQS):

1. Which scale of measurement allows for meaningful ratio comparisons (e.g., "twice as much")?
 - A) Nominal
 - B) Ordinal
 - C) Interval
 - D) Ratio — ANSWER

2. Which stage of analytics answers the question "What will happen?"
 - A) Descriptive
 - B) Diagnostic
 - C) Predictive — ANSWER
 - D) Prescriptive

3. The CRISP-DM phase that focuses on cleaning and transforming raw data is:
 - A) Business Understanding
 - B) Modelling
 - C) Data Preparation — ANSWER
 - D) Evaluation

4. Which sampling method divides the population into subgroups before randomly sampling from each?
 - A) Systematic
 - B) Cluster
 - C) Convenience
 - D) Stratified — ANSWER

5. Business analytics is best described as:
- A) The automated reporting of financial data
 - B) The use of statistics and quantitative methods to improve decision-making
— ANSWER
 - C) The collection of customer feedback surveys
 - D) The installation of ERP software

Questions for Discussion

1. Critically evaluate the statement: "Most Malaysian SMEs do not need business analytics — common sense and experience are sufficient for good decisions." Draw on the analytics maturity model and at least one Malaysian example in your response. (CLO3)
2. Apply the five stages of the analytics lifecycle to a specific business challenge faced by a Sarawak-based organisation of your choice. For each stage, describe the analytics question, the data required, and the expected managerial output. (CLO1)
3. Compare the CRISP-DM framework with a traditional linear project management approach. What are the advantages of CRISP-DM for analytics projects in the context of a medium-sized Malaysian organisation? (CLO2)

REFERENCES

1. **Lind, D. A., Marchal, W. G., & Wathen, S. A.** (2021). *Basic Statistics for Business and Economics* (10th ed.). McGraw-Hill Education.
2. **Evans, J. R.** (2020). *Business Analytics* (3rd ed.). Pearson.
3. **Davenport, T. H., & Harris, J. G.** (2017). *Competing on Analytics: Updated, with a New Introduction* (2nd ed.). Harvard Business Review Press.
4. **Sarawak Digital Economy Corporation (SDEC)** (2023). *Sarawak Digital Economy Blueprint 2030: Progress Report* (Annual Report). SDEC.

Online Learning Resources

The following free resources supplement your study of this unit. All are free or freemium.

IBM: What is Data Analytics? (YouTube, 5 min)

<https://www.youtube.com/watch?v=yZvFH7B6gKI>

**CRISP-DM Framework Explained — Step by Step
(YouTube, 8 min)**

<https://www.youtube.com/watch?v=dCHSd1FzMgo>

**Genspark AI — Business Research & Data Synthesis
Tool (Free)**

<https://www.genspark.ai>

**Julius AI — Upload CSV and ask questions in plain
English (Freemium, 25 free analyses/month)**

<https://julius.ai>

END OF TOPIC

TOPIC

2

○○○○

EXPLORATORY DATA ANALYTICS

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LEARNING OUTCOMES

By the end of this topic you should be able to:

- Calculate and interpret measures of central tendency and position
- Apply numerical descriptive measures to summarise and interpret business datasets
- Design and evaluate dashboards and performance scorecards for managerial use
- Conduct basic exploratory data analysis to generate managerial insights



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LO Learning Objectives

By the end of this topic you should be able to:

- **Calculate and interpret key numerical descriptive measures** including mean, median, and standard deviation.
- **Design performance dashboards and scorecards** to communicate business insights effectively.
- **Apply exploratory analytics techniques** to identify patterns and trends in business datasets.
- **Evaluate descriptive analytics outputs** and translate them into actionable business recommendations.

1 Overview of Exploratory Data Analysis

Exploratory Data Analysis (EDA) is the critical first step in any analytics project. Before applying any formal statistical models, analysts must invest time in understanding the structure, distribution, patterns, and quality of the data they are working with. EDA allows the analyst to see what the data says before imposing assumptions — it is fundamentally a discovery process driven by curiosity rather than hypothesis confirmation.

Goals of Exploratory Data Analysis:

- Detect outliers and anomalies that may distort analysis or signal important events
- Understand variable distributions — whether data is symmetric, skewed, or multi-modal
- Identify relationships between variables — which factors move together or oppose each other
- Check data quality — identify missing values, inconsistencies, and data entry errors
- Generate hypotheses — EDA surfaces patterns that guide subsequent confirmatory analysis

Common EDA Tools and Techniques:

- Frequency tables and bar charts — for categorical variable distributions
- Histograms — for continuous variable distribution shape and spread
- Box plots — for distribution spread, quartiles, and outlier identification
- Scatter plots — for visualising relationships between two continuous variables
- Cross-tabulations (pivot tables) — for relationships between categorical variables
- Summary statistics tables — mean, median, standard deviation, min/max per variable

The managerial value of EDA cannot be overstated. It prevents the "garbage in, garbage out" problem that plagues many analytics initiatives. When managers rush directly to modelling without proper data exploration, they risk building sophisticated models on flawed foundations — producing predictions or recommendations that are confidently wrong. EDA is the quality gate that ensures subsequent analytical work is trustworthy.

2 Measures of Central Tendency

Measures of central tendency describe the "typical" or "average" value in a dataset. Three measures are commonly used: the mean, the median, and the mode. Choosing the appropriate measure depends on the nature of the data and the presence of outliers.

Mean (Arithmetic Average)

The mean is calculated as the sum of all values divided by the number of observations: $\mu = \sum x_i / n$. The mean is most appropriate when data follows a symmetric distribution without extreme outliers. It is sensitive to extreme values — a single very large or very small observation can substantially pull the mean in its direction, making it a misleading representation of the typical value.

Median

The median is the middle value when data is sorted in ascending order. For an even number of observations, it is the average of the two middle values. The median is resistant to outliers, making it the preferred measure of central tendency for skewed distributions.

Malaysian household income data is a classic example — because income distributions are heavily right-skewed (a small number of very high earners pull the mean upward), the Department of Statistics Malaysia reports both the mean and median household income, with the median providing a more representative picture of typical Malaysian household income.

Mode

The mode is the value that appears most frequently in a dataset. It is the only measure of central tendency applicable to nominal (categorical) data. A dataset may have no mode, one mode (unimodal), two modes (bimodal), or multiple modes (multimodal). The mode is particularly useful in retail and marketing contexts — for example, identifying the most commonly purchased product size or the most frequent customer visit day.

Worked Example: Sarawak Logistics Delivery Times

A Sarawak logistics company records delivery times (in hours) for 10 recent shipments from Kuching to Sibul: 24, 28, 22, 35, 26, 28, 30, 24, 22, 41.

Step 1 — Calculate the Mean:

Mean = $(24 + 28 + 22 + 35 + 26 + 28 + 30 + 24 + 22 + 41) / 10 = 280 / 10 = 28.0$ hours

Step 2 — Calculate the Median:

Sorted values: 22, 22, 24, 24, 26, 28, 28, 30, 35, 41

With 10 observations, the median = average of the 5th and 6th values = $(26 + 28) / 2 = 27.0$ hours

Step 3 — Identify the Mode:

Values 22 and 28 each appear twice — the dataset is bimodal: Mode = 22 and 28 hours

Managerial Interpretation:

The mean (28.0 hours) is slightly above the median (27.0 hours), suggesting a mild positive (right) skew in the distribution. This skew is driven by the outlier observation of 41 hours — likely a shipment that encountered an unusual delay. The logistics manager should investigate the 41-hour delivery to identify the cause (weather disruption, route diversion, documentation issue) and determine whether it represents a systemic risk or an isolated event. Reporting only the mean would overstate typical delivery performance.

Self-Reflection



1. Think of a dataset from your own organisation (e.g., sales figures, call handling times, customer satisfaction scores). Which measure of central tendency would be most appropriate to represent "typical" performance? Why? What distortions might arise from using the mean if outliers are present?

3 Measures of Dispersion

While measures of central tendency tell us where the data is centred, measures of dispersion tell us how spread out the data is around that centre. Two datasets can have identical means while having very different distributions — one tightly clustered, the other widely dispersed. For managers, dispersion is critical: high variability in delivery times, production quality, or service performance signals unreliability and operational risk, even if the average appears acceptable.

Range

The range is the simplest measure of dispersion: $\text{Range} = \text{Maximum} - \text{Minimum}$. It is easy to calculate and intuitively understood, but it is highly sensitive to extreme values — a single outlier can dramatically inflate the range, making it an unreliable measure of typical spread.

Variance

Variance measures the average squared deviation of each observation from the mean: $\sigma^2 = \sum (x_i - \bar{x})^2 / (n-1)$. The divisor $(n-1)$ is used for sample variance to produce an unbiased estimator.

Because deviations are squared, variance is expressed in squared units (e.g., hours²), which makes it difficult to interpret directly — this is why standard deviation is typically preferred for communication purposes.

Standard Deviation

The standard deviation is the square root of the variance: $\sigma = \sqrt{[\sum(x_i - \bar{x})^2 / (n-1)]}$. Because it is expressed in the same units as the original data, it is the most widely used and interpreted measure of spread. In a normally distributed dataset, approximately 68% of observations fall within one standard deviation of the mean, and 95% fall within two standard deviations — providing a practical benchmarking framework for operational performance.

Coefficient of Variation (CV)

The Coefficient of Variation standardises the standard deviation as a percentage of the mean: $CV = (\sigma / \mu) \times 100\%$. This makes it possible to compare the relative variability of datasets measured on different scales or in different units — for example, comparing the consistency of delivery time performance across two logistics routes measured in different time zones or different distance ranges.

Interquartile Range (IQR)

The $IQR = Q3 - Q1$, representing the spread of the middle 50% of the data. It is a robust measure of dispersion, unaffected by extreme values, and is the primary measure of spread reported in box plots. The IQR is especially useful when dealing with skewed distributions or data with known outliers.

Worked Example: Delivery Time Dispersion

Using the same delivery time dataset ($\bar{x} = 28.0$ hours): 24, 28, 22, 35, 26, 28, 30, 24, 22, 41.

Deviations from mean and squared deviations:

- $(24 - 28)^2 = 16$
- $(28 - 28)^2 = 0$
- $(22 - 28)^2 = 36$
- $(35 - 28)^2 = 49$
- $(26 - 28)^2 = 4$
- $(28 - 28)^2 = 0$
- $(30 - 28)^2 = 4$
- $(24 - 28)^2 = 16$
- $(22 - 28)^2 = 36$
- $(41 - 28)^2 = 169$

Sum of squared deviations = 330

Sample Variance = $330 / (10 - 1) = 330 / 9 = 36.67$ hours²

Standard Deviation = $\sqrt{36.67} = 6.05$ hours

CV = $(6.05 / 28.0) \times 100\% = 21.6\%$

Managerial Interpretation:

The average delivery time is 28 hours with a standard deviation of approximately 6 hours, indicating significant variability. A CV of 21.6% signals considerable inconsistency in delivery performance — industry benchmarks for express logistics typically target a CV below 10%. The operations manager should investigate the sources of variation: are late deliveries concentrated on specific routes (Kuching–Kapit), specific days of the week, or specific product types (heavy cargo, refrigerated goods)? This would guide targeted process improvement.

Percentiles and Quartiles

Percentiles divide a sorted dataset into 100 equal parts. The p th percentile is the value below which $p\%$ of observations fall. Quartiles are the three values that divide the dataset into four equal parts: Q1 is the 25th percentile (lower quartile), Q2 is the 50th percentile (median), and Q3 is the 75th percentile (upper quartile). Quartiles are widely used in performance benchmarking — for example, a manager may wish to identify which branch outlets fall below the Q1 threshold in customer satisfaction scores for targeted improvement.

Box Plots

A box plot (also known as a box-and-whisker plot) is a standardised visualisation of data distribution. The box spans from Q1 to Q3, representing the IQR and enclosing the middle 50% of the data. A line within the box marks the median (Q2). Whiskers extend from the box to the smallest and largest values that are not outliers — typically defined as values within $1.5 \times \text{IQR}$ of the box boundaries. Data points beyond the whiskers are plotted individually as potential outliers. Box plots are especially powerful for comparing distributions across groups — for example, comparing delivery time distributions across five regional logistics hubs.

Skewness

Skewness measures the asymmetry of a distribution. A positively skewed (right-skewed) distribution has a long tail to the right, meaning a small number of very large values pull the mean above the median — this pattern is characteristic of income distributions in Malaysia, where the mean household income (approximately RM8,479 per month in 2022) exceeds the median (approximately RM6,338 per month), reflecting the influence of high-income households on the national average. A negatively skewed (left-skewed) distribution has a long tail to the left, with the mean falling below the median.

Kurtosis

Kurtosis measures the "peakedness" of a distribution relative to a normal distribution. A leptokurtic distribution (kurtosis > 3) has a sharper peak and heavier tails than the normal distribution — meaning more extreme values occur than expected. A platykurtic distribution (kurtosis < 3) is flatter with lighter tails. For managers, high kurtosis in operational performance data (e.g., system response times, financial transaction values) signals elevated tail risk — the occasional occurrence of extreme events — which may be operationally or financially significant.

5 Dashboards and Performance Scorecards

A dashboard is a visual display of key performance indicators (KPIs) and metrics, designed to enable at-a-glance monitoring of organisational performance. Effective dashboards reduce the cognitive burden on managers by transforming large volumes of data into focused, actionable visual signals. The goal is not to display as much data as possible, but to display the right data — the metrics that most directly inform key management decisions — in the most interpretable form.

Types of Dashboards

Dashboard Type	Time Horizon	Audience	Example KPIs
Operational	Real-time / Daily	Frontline managers	Call queue length, machine uptime, daily sales, on-time delivery rate
Tactical	Weekly / Monthly	Middle management	Customer acquisition cost, order fill rate, defect rate, employee utilisation
Strategic	Quarterly / Annual	Senior executives	Revenue growth, EBITDA margin, market share, Net Promoter Score (NPS)

KPI Design Principles:

- Measurable — expressed as a quantifiable metric with a defined formula
- Relevant — directly linked to a strategic objective or operational priority
- Time-bound — reported at a defined frequency aligned with the decision cycle
- Aligned with strategy — derived from the organisational strategy map, not selected arbitrarily

The Balanced Scorecard

Developed by Kaplan and Norton, the Balanced Scorecard provides a structured framework for selecting KPIs across four perspectives that together provide a balanced view of organisational performance. The Financial perspective captures traditional financial metrics (revenue, profitability, return on equity). The Customer perspective measures customer value delivery (satisfaction, retention, market share). The Internal Process perspective focuses on operational excellence (process efficiency, quality, cycle time). The Learning and Growth perspective tracks the organisation's capacity for innovation and development (employee skills, system capabilities, culture). By spanning all four perspectives, the Balanced Scorecard guards against the common managerial tendency to over-emphasise short-term financial metrics at the expense of longer-term value drivers.

Case Study: SDEC Digital Economy Dashboard

The Sarawak Digital Economy Corporation (SDEC) operates a statewide digital economy dashboard tracking key metrics across six priority sectors: digital infrastructure, digital commerce, digital content, cybersecurity, digital government, and digital talent. The dashboard aggregates data from government agencies, telecommunications providers (Telekom Malaysia, Maxis, and Celcom), and private sector partners to provide quarterly progress reports against the Sarawak Digital Economy Blueprint 2030 targets.

Key indicators monitored include broadband coverage (target: 98% by 2025, achieved: 91.2% as of 2023), registered digital businesses, e-commerce gross merchandise value (GMV) generated by Sarawak-based sellers on platforms including Shopee and Lazada, and the number of certified digital talent graduates from programmes under the SDEC Digital Skills Academy.

The dashboard is presented quarterly to the Sarawak State Cabinet and published in the SDEC Annual Report. This exemplifies how descriptive analytics and dashboard design can support strategic monitoring, stakeholder accountability, and policy adjustment at the state government level.

6 Descriptive Analytics Applications

Frequency Tables and Histograms

Frequency tables summarise the distribution of a variable by counting the number of observations that fall into each category or interval. For continuous variables, the data is grouped into class intervals (bins) of equal width. A histogram visualises this frequency table as a bar chart where bar height represents frequency (or relative frequency), providing an immediate visual impression of distribution shape — whether it is symmetric, skewed, bimodal, or uniform. Frequency analysis is a standard first step in understanding any new dataset.

Cross-Tabulation

A cross-tabulation (also called a contingency table or pivot table) displays the joint frequency distribution of two or more categorical variables. It reveals patterns of association between variables — for example, whether customer satisfaction ratings differ across product categories, or whether return rates vary by sales channel. The chi-square test of independence can be applied to a cross-tabulation to formally test whether the observed association between two categorical variables is statistically significant.

Data Profiling

Data profiling is the systematic, automated process of examining a dataset to produce summary statistics and quality reports. A data profiling exercise generates: null counts and missing value percentages per variable; value distributions and frequency counts; minimum, maximum, and range for numeric variables; cardinality (number of unique values) for categorical variables; outlier flags using IQR or z-score methods.

Data profiling is typically the first analytical step when receiving a new dataset, and its findings directly shape the data preparation strategy in the CRISP-DM framework.

Self-Reflection



2. Imagine you are a marketing analyst at a Malaysian e-commerce company similar to Shopee Malaysia. You have access to 3 months of customer transaction data. List three exploratory questions you would investigate and describe the visualisation or descriptive statistic you would use to answer each question.

SUMMARY

EDA is the foundation of any analytics project — it uncovers data quality issues, reveals distribution shapes, and generates hypotheses for further investigation.

Mean, median, and mode each measure central tendency differently; the median is preferred for skewed distributions and the mode for categorical data.

Variance and standard deviation measure data spread around the mean; the Coefficient of Variation allows comparison of relative variability across different scales.

Quartiles, percentiles, skewness, and kurtosis describe the shape and position of distributions, providing a complete picture of data behaviour beyond the average.

Dashboards translate complex data into at-a-glance performance monitoring — operational dashboards serve frontline managers while strategic dashboards serve senior executives.

The Balanced Scorecard provides a structured, multi-perspective framework for KPI design, balancing financial, customer, internal process, and learning dimensions.

Multiple Choice Questions (MCQS):

1. Which measure of central tendency is most resistant to outliers?
 - A) Mean
 - B) Mode
 - C) Median — ANSWER
 - D) Variance

2. The Coefficient of Variation (CV) is calculated as:
 - A) Mean / Standard Deviation
 - B) $(\text{Standard Deviation} / \text{Mean}) \times 100\%$ — ANSWER
 - C) Variance / Mean
 - D) IQR / Median

3. Which type of dashboard is most appropriate for monitoring real-time factory floor performance?
 - A) Strategic
 - B) Analytical
 - C) Operational — ANSWER
 - D) Balanced

4. A dataset with a positive (right) skew has:
 - A) Mean = Median
 - B) Mean < Median
 - C) Mean > Median — ANSWER
 - D) Mode > Mean

5. The IQR (Interquartile Range) is calculated as:

- A) $Q1 - Q3$
- B) $Q3 + Q1$
- C) $Q3 - Q1$ — ANSWER
- D) Maximum – Minimum

Questions for Discussion

1. A regional bank in Sarawak is reviewing its loan processing times. The average processing time is 14 days with a standard deviation of 8 days. Using descriptive statistics, how would you identify whether there is a performance problem, and what further EDA steps would you recommend? Link your analysis to CLO1.
2. Design a strategic dashboard for a Sarawak hospitality company with five hotels across Kuching, Miri, and Sibü. Specify: (a) the four most important KPIs; (b) the appropriate visualisation for each KPI; (c) the data sources required; and (d) how the dashboard aligns with the Balanced Scorecard perspectives. (CLO2)
3. Critically evaluate whether descriptive analytics alone is sufficient for managerial decision-making in a competitive environment. Use at least one Malaysian company example. (CLO3)

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2. **Jaggia, S., & Kelly, A.** (2020). *Business Statistics: Communicating with Numbers* (3rd ed.). McGraw-Hill Education.
3. **Evans, J. R.** (2020). *Business Analytics* (3rd ed.). Pearson.
4. **Few, S.** (2012). *Show Me the Numbers: Designing Tables and Graphs to Enlighten* (2nd ed.). Analytics Press.

Online Learning Resources

The following free resources supplement your study of this unit.

StatQuest: Statistics Fundamentals Playlist (YouTube, Free)

<https://www.youtube.com/playlist?list=PLb1h5JKOoLUK0FLuzwntyYI10UQFUhsY9>

CrashCourse Statistics — Descriptive Statistics (YouTube, 12 min)

https://www.youtube.com/watch?v=uhxtUt_-GyM

Microsoft Copilot in Excel — AI-powered Descriptive Analytics (Free with Microsoft 365)

<https://support.microsoft.com/en-us/copilot-excel>

Google Looker Studio — Free Cloud Dashboard Builder

<https://lookerstudio.google.com>

END OF TOPIC

TOPIC

3

DIAGNOSTIC ANALYTICS

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LEARNING OUTCOMES

By the end of this topic you should be able to:

- **Apply measures of dispersion and hypothesis testing to diagnose business performance issues**
- **Conduct and interpret t-tests, ANOVA, and correlation analyses in managerial contexts**
- **Perform trend analysis to identify patterns and anomalies in business data**
- **Apply diagnostic analytics to identify root causes of business problems in Malaysian organisations**



INSPIRING MINDS, SHAPING FUTURE

LO Learning Objectives

By the end of this topic you should be able to:

- **Apply hypothesis testing methods** including t-tests, ANOVA, and correlation, to business problems.
- **Interpret measures of dispersion** and assess data variability in business contexts.
- **Use diagnostic analytics models** to identify root causes of business performance issues.
- **Conduct trend analysis** to diagnose historical patterns and inform forward-looking decisions.

1 Overview of Diagnostic Analytics

Diagnostic analytics is the second stage of the analytics maturity model, positioned between descriptive analytics (what happened?) and predictive analytics (what will happen?). While descriptive analytics summarises historical performance, diagnostic analytics applies statistical techniques to answer the more managerially valuable question: why did this performance occur? Understanding root causes, rather than merely observing outcomes, is the foundation of effective managerial intervention.

Core Diagnostic Analytics Techniques:

- Hypothesis testing — formally testing whether observed differences or relationships are statistically significant
- t-tests — comparing means between one or two groups to determine if differences are real or due to sampling chance
- ANOVA (Analysis of Variance) — extending mean comparison to three or more groups
- Correlation analysis — measuring the strength and direction of linear relationships between variables
- Simple regression analysis — quantifying the relationship between a dependent and an independent variable
- Trend analysis — examining time-series data for patterns, seasonality, and anomalies

- Root cause analysis — structured problem-solving using fishbone diagrams and 5-Why analysis informed by data

The managerial value of diagnostic analytics is profound. Without it, managers risk misdiagnosis — allocating resources to treat visible symptoms rather than underlying causes. For example, a retail chain experiencing declining same-store sales may attribute the problem to pricing (a symptom) when the actual root cause is a deteriorating customer experience caused by understaffing (a cause). Diagnostic analytics provides the evidence to distinguish symptom from cause, directing management action where it will have the greatest impact.

2 Hypothesis Testing

Hypothesis testing is the formal statistical framework for making evidence-based decisions about population parameters from sample data. It provides a structured, reproducible procedure for determining whether observed patterns in data are genuine features of the underlying reality or merely the result of random sampling variation.

The Logic of Hypothesis Testing

Every hypothesis test begins by stating two competing claims. The null hypothesis (H_0) is the default position — typically a statement of "no effect" or "no difference" that will be maintained unless the data provides sufficient evidence against it. The alternative hypothesis (H_1) is the research claim — the assertion that a significant effect or difference exists. The analyst then collects sample data and calculates a test statistic that measures how far the sample result deviates from what would be expected if H_0 were true. If this deviation is large enough (quantified through the p-value), H_0 is rejected in favour of H_1 .

Type I and Type II Errors

Error Type	Description	Consequence	Controlled By
Type I Error (α)	Rejecting H_0 when it is actually true — a false positive	Acting on a non-existent effect; wasted resources	Setting significance level α (typically 0.05)
Type II Error (β)	Failing to reject H_0 when it is actually false — a false negative	Missing a real effect; failing to act on a genuine problem	Increasing sample size; increasing statistical power

The Five-Step Hypothesis Testing Process:

1. State H_0 and H_1 clearly, specifying the direction of the test (one-tail or two-tail) if applicable
2. Choose the significance level α — typically 0.05 (5%), though 0.01 may be used for high-stakes decisions
3. Select the appropriate test statistic based on the data type, sample size, and distributional assumptions
4. Calculate the test statistic and corresponding p-value from the sample data
5. Make the decision: if $p < \alpha$, reject H_0 (sufficient evidence for H_1); if $p \geq \alpha$, fail to reject H_0

Interpreting the p-value:

The p-value is the probability of observing the sample data (or data more extreme) if the null hypothesis were true. A small p-value (e.g., $p = 0.02$) indicates that the observed result would be very unlikely under H_0 , providing evidence against it. Critically, the p-value is NOT the probability that H_0 is true — this is a common and consequential misinterpretation. A p-value of 0.03 means "if H_0 were true, there is only a 3% chance of seeing data this extreme or more extreme by chance alone."

One-Tail vs Two-Tail Tests:

A two-tail test is used when the alternative hypothesis does not specify a direction — the analyst tests whether the parameter differs from the null value in either direction ($H_1: \mu \neq \mu_0$). A one-tail test is appropriate when theory or prior evidence specifies the expected direction of the effect — for example, testing whether a new process reduces (not just changes) average cycle time ($H_1: \mu < \mu_0$). One-tail tests have greater statistical power for detecting effects in the specified direction but should only be used when the directional prediction is made before data collection.

Worked Example: Sarawak Timber Company — Drying Process

A Sarawak timber processing company has implemented a new kiln drying process and wants to statistically verify whether it reduces average board moisture content below the industry standard of 15%. A quality engineer samples 25 boards from the new process and measures their moisture content, obtaining a sample mean of $\bar{x} = 13.8\%$ and sample standard deviation of $s = 2.1\%$.

Setting up the test:

$H_0: \mu \geq 15\%$ (the new process does NOT reduce moisture below the standard)

$H_1: \mu < 15\%$ (the new process DOES reduce moisture below the standard)

$\alpha = 0.05$; one-tail test (left-tail, since we expect a reduction)

Calculating the t-statistic:

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n}) = (13.8 - 15) / (2.1 / \sqrt{25}) = -1.2 / (2.1 / 5) = -1.2 / 0.42 = -2.857$$

Degrees of freedom: $df = n - 1 = 24$

Critical value (one-tail, $\alpha = 0.05$, $df = 24$): $t_{\text{critical}} = -1.711$

Decision:

Since the calculated t-statistic (-2.857) is less than the critical value (-1.711), we fall in the rejection region. We reject H_0 at the 5% significance level. Conclusion: There is sufficient statistical evidence to conclude that the new drying process significantly reduces average board moisture content below the 15% standard. The timber company can proceed with full implementation of the new process with statistical confidence.

3**t-tests**

The t-test is a family of statistical tests used to compare means. The appropriate variant depends on the number of groups being compared and the research design. All t-tests assume that the variable of interest is approximately normally distributed (or that the sample is large enough for the Central Limit Theorem to apply) and that observations are independent.

One-Sample t-Test

Used to test whether a sample mean differs significantly from a known or hypothesised population value. The test statistic is: $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$. This is the test applied in the Sarawak timber company example above. Common managerial applications include testing whether a process average meets a regulatory standard, or whether a product specification is being consistently met.

Independent Samples t-Test

Used to compare the means of two independent groups — groups with no overlap in membership. This test is applied when a manager wants to know whether performance differs between two distinct groups, such as two branches, two product lines, or two customer segments. The test assumes that the two group variances are approximately equal (this can be checked using Levene's test for equality of variances; if violated, a Welch's t-test correction is applied). A Malaysian HR analytics application would involve comparing average productivity scores (sales per day) between KL-based and Kuching-based branches of a bank, to determine whether the observed difference in branch performance is statistically significant or attributable to sampling variation.

Paired Samples t-Test

Used to compare two measurements taken from the same subjects or matched pairs — for example, before and after a training programme, or measuring the same KPI under two different conditions. By pairing observations, this test removes between-subject variability that would otherwise inflate the standard error, making it more powerful than the independent samples test when repeated measurements are available. It is particularly appropriate for pre/post programme evaluations in HR and operations management.

Key Assumptions for All t-Tests:

- Normality — the variable is approximately normally distributed (or $n > 30$ for CLT)
- Independence — observations are independent of each other
- Equal variance — for independent samples t-test (check with Levene's test)

Self-Reflection



1. Your organisation has just introduced a new training programme. Before the programme, average monthly sales per employee was RM45,000. Three months later, a sample of 20 trained employees shows average sales of RM48,500 ($s = \text{RM}6,200$). How would you set up a hypothesis test to determine if the training significantly improved performance? What would rejection of H_0 mean for management decisions?

4

ANOVA — Analysis of Variance

One-Way Analysis of Variance (ANOVA) extends the logic of the t-test to situations where three or more group means are being compared simultaneously. While it would be tempting to conduct multiple pairwise t-tests (e.g., Group A vs B, A vs C, B vs C), doing so inflates the probability of making at least one Type I error — this is known as the multiple comparisons problem. ANOVA addresses this by testing all group means in a single test while maintaining the specified significance level α .

The F-Statistic

ANOVA partitions the total variability in the data into two components: between-group variance (variation attributable to differences between group means) and within-group variance (random variation within each group). The F-statistic is the ratio of these two variance estimates: $F = MS(\text{Between}) / MS(\text{Within})$. A large F-statistic (and corresponding small p-value) indicates that the between-group variance is substantially larger than the within-group variance, providing evidence that at least one group mean differs significantly from the others.

The ANOVA Table Structure:

Source	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Statistic	p-value
Between Groups	SS(B)	$k - 1$	$MS(B) = SS(B)/(k-1)$	$F = MS(B)/MS(W)$	From F-distribution
Within Groups	SS(W)	$n - k$	$MS(W) = SS(W)/(n-k)$	—	—
Total	SS(T)	$n - 1$	—	—	—

Where k = number of groups and n = total sample size across all groups.

Post-Hoc Testing: Tukey HSD

A significant ANOVA result tells us only that at least one group mean is different — it does not identify which pairs of groups differ. Post-hoc tests are required for this purpose. The Tukey Honestly Significant Difference (Tukey HSD) test is the most widely used post-hoc procedure. It performs all pairwise comparisons between group means while controlling the family-wise error rate, providing adjusted p-values and confidence intervals for each pair.

Worked Example: Malaysian Retailer — City Performance Comparison

A Malaysian retail chain with outlets in Kuching, Miri, and Kota Kinabalu (KK) wants to determine whether average weekly sales differ significantly across the three cities. A sample of 15 weekly sales figures per city is collected. The one-way ANOVA is conducted in Microsoft Excel or SPSS, yielding the following output:

Source	SS	df	MS	F	p-value
Between Groups	2,847,320	2	1,423,660	5.73	0.009
Within Groups	10,450,180	42	248,814	—	—
Total	13,297,500	44	—	—	—

Decision and Post-Hoc Analysis:

Since $F = 5.73$ and $p = 0.009 < \alpha = 0.05$, we reject H_0 . At least one city's average weekly sales differ significantly from the others. Tukey HSD post-hoc analysis reveals the following pairwise comparisons:

- Kuching vs Miri: Mean difference = RM4,820; $p = 0.012 \rightarrow$ Statistically significant
- Kuching vs KK: Mean difference = RM2,940; $p = 0.087 \rightarrow$ Not statistically significant
- Miri vs KK: Mean difference = RM1,880; $p = 0.341 \rightarrow$ Not statistically significant

Managerial Interpretation:

The statistically significant difference is specifically between Kuching and Miri — with Kuching significantly outperforming Miri on average weekly sales. Management should investigate the drivers of this difference. Potential diagnostic hypotheses include: differences in location catchment area size and demographics; differences in store staffing levels and customer service quality; differences in local marketing investment and promotional activity; differences in product assortment alignment with local consumer preferences; and differences in competitive environment (e.g., proximity of competing retailers or shopping malls in each city).

Correlation analysis measures the strength and direction of the linear relationship between two continuous variables. The Pearson correlation coefficient r is the most widely used measure, ranging from -1.00 (perfect negative linear relationship) to $+1.00$ (perfect positive linear relationship), with 0 indicating no linear relationship. The formula is: $r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{[\sum(x_i - \bar{x})^2 \times \sum(y_i - \bar{y})^2]}}$.

Interpreting Correlation Strength:

$ r $ Range	Interpretation
0.00 – 0.19	Very weak — negligible linear relationship
0.20 – 0.39	Weak — slight linear tendency
0.40 – 0.59	Moderate — meaningful but not strong linear relationship
0.60 – 0.79	Strong — substantial linear relationship
0.80 – 1.00	Very strong — near-linear relationship

Correlation Does NOT Equal Causation

The most important caution in correlation analysis is that a strong statistical correlation between two variables does not imply that one causes the other. Both variables may be driven by a common third factor (confounding), or the correlation may be entirely coincidental (spurious). A classic spurious correlation example: ice cream sales and drowning incidents are strongly positively correlated — but ice cream does not cause drowning. Both are driven by the confounding variable of hot weather (more people eat ice cream AND more people swim in hot weather). In a Malaysian business context, a company might observe a strong positive correlation between its advertising spend and competitor revenue — this is spurious and would lead to entirely erroneous management decisions if treated as causal.

Spearman Rank Correlation

When data is ordinal, or when the distribution of continuous variables is heavily skewed or non-normal, the Spearman rank correlation coefficient (r_s) is preferred over Pearson r . Spearman correlation operates on the ranks of the data rather than the raw values, making it robust to outliers and non-normality. It is interpreted in the same direction and magnitude as Pearson r .

Worked Example: Sarawak Tourism Correlation

A tourism analyst examines the relationship between monthly international tourist arrivals to Sarawak and the monthly hotel occupancy rate across 36 months of data. The Pearson correlation coefficient is calculated as $r = 0.84$.

Interpretation: $r = 0.84$ indicates a very strong positive linear correlation between tourist arrivals and hotel occupancy rates. As tourist arrivals increase, hotel occupancy rates increase proportionately. This finding is statistically significant ($p < 0.001$ at $n = 36$). Managerial implication: Sarawak Tourism Board and the Sarawak Hotel Association can use tourist arrival forecasts as a leading indicator for occupancy rate planning, allowing hotels to calibrate staffing, pricing, and inventory decisions in advance of anticipated demand peaks.

Correlation Matrix for Multiple Variables:

When multiple variables are available, a correlation matrix displays all pairwise correlation coefficients in a symmetric table. This provides a rapid overview of the inter-relationships in the dataset, helping analysts identify which variable pairs to investigate further and which independent variables may be collinear (strongly correlated with each other) — an important consideration for subsequent regression modelling.

Trend analysis examines a variable measured over time — a time series — to identify systematic patterns, directional movements, and anomalies. Understanding the temporal behaviour of business metrics is fundamental to diagnosis: was the observed performance decline a one-time event, a seasonal dip, or the beginning of a structural downward trend? Only trend analysis can answer this question.

Types of Patterns in Time-Series Data:

- Trend — a long-term directional movement (upward growth, structural decline, or stagnation)
- Seasonality — regular, repeating patterns tied to the calendar (monthly, quarterly, annual cycles)
- Cyclical — longer-term fluctuations associated with economic or industry cycles (multi-year patterns)
- Irregular/Residual — random, unexplained variation after accounting for trend, seasonality, and cyclical components

Moving Averages

A simple moving average (SMA) calculates the average of a specified number of consecutive observations, "rolling" forward one period at a time. By averaging out short-term fluctuations, the moving average reveals the underlying trend direction more clearly. A weighted moving average (WMA) assigns greater weight to more recent observations, making the average more responsive to recent changes. Moving averages are widely used in retail (smoothing daily sales to see weekly trends), finance (smoothing stock prices), and operations (smoothing production output data).

Linear Trend Line

A linear trend line fits a straight line to the time-series data using least-squares regression: $y = a + bx$, where y is the variable of interest, x is the time period index, a is the y -intercept, and b is the slope. The slope b represents the average change in y per unit of time — for example, $b = 1,250$ would indicate average growth of 1,250 units per period. A positive slope confirms an upward trend; a negative slope confirms a downward trend. The strength of the linear trend is assessed using R^2 — the proportion of total variation explained by the linear trend component.

Time-Series Decomposition

Decomposition separates a time series into its individual components: Trend (T), Seasonality (S), and Residual/Irregular (R). In an additive decomposition model: $y = T + S + R$. In a multiplicative model: $y = T \times S \times R$. Multiplicative decomposition is more appropriate when seasonal fluctuations grow proportionally with the trend level. Decomposition enables managers to separate "what was expected" (trend + seasonality) from "what was unexpected" (residual) — the residual component is where diagnostic investigation should be focused when anomalies are observed.

Case Study: Sarawak Tourism Diagnostic Analytics

From 2018 to 2024, Sarawak's tourism arrival data revealed a complex pattern that required diagnostic analytics to fully interpret. The historical trend showed a strong upward movement from 4.8 million arrivals in 2018 to 5.1 million in 2019, followed by a catastrophic COVID-19-driven collapse to approximately 0.9 million in 2020 and 0.7 million in 2021 as international borders closed and domestic movement was restricted. A V-shaped recovery commenced in 2022 (2.1 million arrivals) and accelerated through 2023 (4.4 million), culminating in a record 5.6 million arrivals in 2024.

Diagnostic analytics was applied to identify the key drivers of the 2024 record performance. Correlation analysis across 36 months of recovery data revealed strong positive associations between tourist arrivals and three leading indicators: the number of direct flight routes from Singapore and Kuala Lumpur to Kuching and Miri airports ($r = 0.79$, strong positive correlation); the Sarawak Tourism Board's Visit Sarawak Campaign digital marketing spend ($r = 0.83$, very strong positive correlation); and CNY and Gawai festival periods (confirmed as seasonal peaks through decomposition analysis).

One-way ANOVA across five tourism zones — Kuching, Miri, Sibul, Limbang, and Ulu Sarawak — showed significantly different recovery trajectories ($F = 7.21$, $p = 0.001$). Post-hoc Tukey HSD testing identified that Kuching and Miri recovered significantly faster than Sibul, Limbang, and Ulu Sarawak. Diagnostic analysis attributed this primarily to the international air connectivity advantage of Kuching International Airport and Miri Airport, which direct flight service from Singapore, Jakarta, and Brunei resumed in 2022, compared to interior zones dependent on domestic connectivity only. This finding directly informed the Sarawak Tourism Board's strategic decision to prioritise investment in regional air route development and digital marketing for the 2025–2026 campaign cycle.

Self-Reflection



2. Access any time-series data from your organisation (e.g., monthly revenue, quarterly production output, weekly customer complaints). Identify: (a) the overall trend direction; (b) any seasonal patterns; (c) any anomalies or unexplained spikes/drops. What diagnostic hypotheses would you form to explain the anomalies?

SUMMARY

Diagnostic analytics moves beyond description to identify the causes of observed patterns — turning "what happened" into "why it happened" and guiding targeted managerial intervention.

Hypothesis testing provides a structured statistical framework for making evidence-based decisions, with clear rules for rejecting or maintaining the null hypothesis based on the p-value and significance level.

t-tests compare means for one sample or two groups; independent samples t-test compares two distinct groups while paired t-test compares pre/post measurements on the same subjects.

ANOVA extends mean comparison to three or more groups using the F-statistic; post-hoc Tukey HSD testing identifies which specific group pairs differ significantly.

Pearson correlation measures the strength and direction of linear relationships between variables; always remember that correlation does not imply causation.

Trend analysis decomposes time-series data into trend, seasonal, cyclical, and residual components, enabling managers to distinguish expected patterns from genuine anomalies requiring diagnostic investigation.

Multiple Choice Questions (MCQS):

1. In hypothesis testing, a Type I error occurs when:
 - A) We fail to reject H_0 when it is false
 - B) We reject H_0 when it is true — ANSWER
 - C) The p-value exceeds 0.05
 - D) The sample size is too small

2. ANOVA is used to compare:
 - A) Two sample proportions
 - B) A sample mean to a population mean
 - C) Means across three or more groups — ANSWER
 - D) Two sample variances

3. A Pearson correlation of $r = -0.82$ indicates:
 - A) Weak positive relationship
 - B) No relationship
 - C) Strong negative relationship — ANSWER
 - D) Moderate positive relationship

4. The p-value in a hypothesis test represents:
 - A) The probability that H_0 is true
 - B) The probability of the observed data (or more extreme) given H_0 is true — ANSWER
 - C) The significance level α
 - D) The confidence interval width

5. Which post-hoc test is commonly used after a significant one-way ANOVA to identify which group pairs differ?

- A) Bonferroni
- B) Durbin-Watson
- C) Tukey HSD — ANSWER
- D) Levene's

Questions for Discussion

1. A Malaysian logistics company notices that delivery times increased by an average of 3.2 hours in Q3 2024 compared to Q3 2023. Design a full diagnostic analytics investigation: state the hypotheses, select appropriate statistical tests, describe the data required, and explain how you would interpret the results to management. (CLO1)
2. Critically discuss the risks of using correlation analysis alone to draw causal conclusions in business analytics. Provide two examples from the Malaysian business context where spurious correlations could lead to poor management decisions. (CLO3)
3. Apply trend analysis to a Sarawak industry of your choice. Identify what historical patterns the data would likely reveal and what diagnostic questions would most benefit management in that industry. (CLO2)

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3. **Evans, J. R.** (2020). *Business Analytics* (3rd ed.). Pearson.
4. **Waller, M. A., & Fawcett, S. E.** (2013). *Data Science, Predictive Analytics, and Big Data: A Revolution That Will Transform Supply Chain Design and Management* (Journal of Business Logistics, 34(2)). Wiley.

Online Learning Resources

The following free resources supplement your study of this unit.

**CrashCourse Statistics — Hypothesis Testing
(YouTube, 11 min)**

<https://www.youtube.com/watch?v=bf3egy7TQ2Q>

StatQuest: ANOVA Explained Clearly (YouTube, 18 min)

<https://www.youtube.com/watch?v=0Vj2V2qRU10>

JASP — Free Point-and-Click Statistics Software (No Coding Required)

<https://jasp-stats.org>

Julius AI — Run t-tests and ANOVA by uploading your Excel data

<https://julius.ai>

END OF TOPIC

TOPIC

4

PREDICTIVE ANALYTICS

*Lecturers: Dr. Colin Wong Koh King; Faizul Bin Ayupp
Dino; AP Michael Tinggi*

LEARNING OUTCOMES

By the end of this topic you should be able to:

- **Explain the key concepts and techniques of predictive analytics for business applications**
- **Build and interpret regression models to forecast business outcomes**
- **Apply classification and clustering techniques to segment and predict customer behaviour**
- **Evaluate the performance of predictive models using appropriate metrics**



INSPIRING MINDS, SHAPING FUTURE

LO Learning Objectives

By the end of this topic you should be able to:

- **Apply regression and classification techniques** to build predictive models for business forecasting.
- **Use clustering algorithms** to identify market segments and customer groupings.
- **Implement association rule mining** to uncover patterns in transactional business data.
- **Evaluate the accuracy and applicability of predictive models** for managerial decision support.

1 Introduction to Predictive Analytics

Predictive analytics uses historical data, statistical algorithms, and machine learning to forecast future outcomes or classify new observations. It answers the question "What is likely to happen?" — enabling organisations to act proactively rather than reactively.

Predictive analytics techniques fall into two broad categories:

- **Supervised learning:** the model is trained on labelled data (where the outcome is known) and learns to predict the outcome for new, unseen data. Examples: regression, classification.
- **Unsupervised learning:** the model finds structure in unlabelled data without a predefined outcome variable. Examples: clustering, association rules.

Key predictive techniques are summarised in the table below:

Technique	Type	Output	Common Use Case
Linear Regression	Supervised	Continuous value	Sales forecasting, price estimation
Logistic Regression	Supervised	Binary category	Churn prediction, loan default
Decision Trees	Supervised	Category or value	Credit scoring, fraud detection
Random Forest	Supervised	Category or value	Risk classification
K-Means Clustering	Unsupervised	Cluster membership	Customer segmentation
Association Rules	Unsupervised	Item associations	Market basket analysis

Model Validation: Training Data, Test Data, and the Train/Test Split

Before deploying a predictive model, analysts split the available data into a training set (typically 80%) used to fit the model, and a test set (typically 20%) used to evaluate how well the model generalises to new data. This prevents overfitting — where the model "memorises" the training data but fails on new observations.

Overfitting vs Underfitting:

- Overfitting: the model is too complex and captures noise rather than the true signal. Analogy: a student who memorises past exam answers but cannot answer a new question.
- Underfitting: the model is too simple and fails to capture the underlying pattern. Analogy: a student who barely studied and cannot answer any question.
- The goal is a model with the right level of complexity — good performance on both training and test data.

Model Performance Metrics Overview:

- Classification models: Accuracy, Precision, Recall (Sensitivity), F1-Score, ROC-AUC.
- Regression models: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R-squared (R^2).

2 Regression Analysis

Simple Linear Regression

Simple linear regression models the relationship between one predictor variable (X) and one continuous outcome variable (Y) using a straight line:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- β_0 : the intercept — the value of Y when $X = 0$.
- β_1 : the slope — the change in Y for each one-unit increase in X.
- ε : the error term — representing the variation in Y not explained by X.

Interpreting Regression Output:

- Coefficient (β_1): size and direction of the effect of X on Y.
- p-value: tests whether the coefficient is statistically significantly different from zero. A p-value < 0.05 typically indicates the predictor is statistically significant.
- R^2 (R-squared): the proportion of variance in Y explained by the model. $R^2 = 0.76$ means 76% of the variation in Y is explained by the predictors.
- F-statistic: tests whether the model as a whole is statistically significant.

Worked Example: Sarawak Energy Load Forecasting

Context: Sarawak Energy Berhad (SEB) wants to predict daily peak electricity demand (MW) based on daily maximum temperature ($^{\circ}\text{C}$). Using one year of historical data, the following regression output was obtained:

- Regression Equation: Peak Demand = $450.2 + 18.6 \times \text{Temperature}$
- $R^2 = 0.76$
- F-statistic = 87.4 ($p < 0.001$)
- Temperature coefficient p-value < 0.001

Interpretation: For every 1°C increase in maximum daily temperature, peak electricity demand increases by 18.6 MW. The model explains 76% of the variation in peak demand. The overall model and the temperature coefficient are both statistically significant ($p < 0.001$).

Managerial Implication: SEB can use daily weather forecasts to anticipate peak electricity demand and pre-position generation capacity accordingly, reducing the risk of supply shortfalls and costly emergency procurement.

Multiple Linear Regression

Multiple linear regression extends the model to include several predictor variables: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$. Adding more predictors generally increases R^2 , but the adjusted R^2 penalises for adding irrelevant predictors and is a more reliable measure of model quality.

Analysts must also check for multicollinearity — when two or more predictors are highly correlated with each other — which can make individual coefficients unreliable.

Self-Reflection



1. Think of a business outcome in your organisation that you would like to predict (e.g., monthly sales, customer churn rate, project completion time). What predictor variables would you include in a regression model? How would you validate the model before relying on it for decisions?

Logistic Regression

Logistic regression predicts the probability that an observation belongs to a binary category (e.g., Yes/No, Churn/No Churn). The output is a probability between 0 and 1. A threshold (commonly 0.5) is used to classify observations: if predicted probability > 0.5 , classify as "Yes"; otherwise classify as "No". The threshold can be adjusted depending on the cost of false positives versus false negatives.

Decision Trees

A decision tree classifies observations by recursively splitting the data based on feature values, forming a tree-like structure. Each internal node represents a decision rule on a feature, and each leaf node represents a predicted class. Decision trees are highly interpretable and can be easily explained to management — a significant advantage in regulated industries such as banking and insurance.

Confusion Matrix

The confusion matrix summarises the performance of a classification model on a test set:

- True Positive (TP): model correctly predicted "Yes" and actual is "Yes".
- True Negative (TN): model correctly predicted "No" and actual is "No".
- False Positive (FP): model incorrectly predicted "Yes" but actual is "No" (Type I error).
- False Negative (FN): model incorrectly predicted "No" but actual is "Yes" (Type II error).

Performance Metrics:

- Accuracy = $(TP + TN) / \text{Total}$ — proportion of all predictions that are correct.
- Precision = $TP / (TP + FP)$ — of all predicted positives, what proportion is actually positive?
- Recall (Sensitivity) = $TP / (TP + FN)$ — of all actual positives, what proportion did the model identify?

- $F1\text{-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$ — the harmonic mean of precision and recall, useful when class imbalance exists.

Worked Example: Malaysian Bank Customer Churn Prediction

A Malaysian bank built a logistic regression model to predict which customers are likely to close their accounts within 90 days. Training data: 10,000 customers with 12 features including age, account tenure, transaction frequency, number of products held, average balance, days since last login, and complaint history.

Confusion matrix on the test set ($n = 2,000$ customers):

- True Positives (TP) = 180 (churners correctly identified)
- True Negatives (TN) = 1,620 (retained customers correctly identified)
- False Positives (FP) = 80 (retained customers incorrectly flagged as churners)
- False Negatives (FN) = 120 (actual churners missed by the model)
- Accuracy = $(180 + 1,620) / 2,000 = 90.0\%$
- Precision = $180 / (180 + 80) = 180 / 260 = 69.2\%$
- Recall = $180 / (180 + 120) = 180 / 300 = 60.0\%$

Managerial Interpretation: The model correctly identifies 60% of actual churners. However, 40% of actual churners are missed ($FN = 120$). In a churn management context, missing a churner (FN) is typically more costly than incorrectly flagging a retained customer (FP). Management may choose to lower the classification threshold (e.g., from 0.5 to 0.3) to increase recall — catching more churners — at the cost of slightly lower precision.

Case: PETROS Customer Analytics

PETROS, Sarawak's state petroleum company, uses classification analytics to triage supplier applications. Suppliers are classified into high-risk, medium-risk, and low-risk categories based on financial health indicators, regulatory compliance history, and operational capacity data. The classification model reduced manual supplier vetting time by 45% and improved the accuracy of high-risk supplier identification by 30%, enabling the procurement team to focus manual review efforts on the highest-risk applications.

4 Clustering

K-Means Clustering

K-Means is the most widely used clustering algorithm. It partitions observations into k clusters, where each observation belongs to the cluster with the nearest centroid (mean). The algorithm proceeds as follows:

1. Choose the number of clusters k .
2. Randomly initialise k cluster centroids in the feature space.
3. Assign each observation to the nearest centroid (using Euclidean distance).
4. Recalculate the centroid of each cluster as the mean of all assigned observations.
5. Repeat steps 3–4 until centroid positions stabilise (convergence).

Choosing the Optimal Number of Clusters — The Elbow Method:

Run K-Means for $k = 1, 2, 3, \dots, n$ and calculate the Within-Cluster Sum of Squares (WCSS) for each k . Plot WCSS against k . The "elbow" — the point where adding another cluster produces diminishing reductions in WCSS — indicates the optimal k .

Interpreting Clusters:

After clustering, profile each cluster by calculating the mean values of all key variables for observations in that cluster. Label each cluster descriptively based on its distinguishing characteristics. These labels help managers understand and act on the segmentation.

Worked Example: Sarawak SME Segmentation

The Sarawak Digital Economy Corporation (SDEC) analysed 500 Sarawak SMEs across three variables: (1) digital adoption score (0–100), (2) annual revenue growth (%), and (3) employee count. K-Means with $k = 3$ produced three distinct clusters:

- Cluster 1 — "Digital Leaders" ($n = 85$): Mean digital score = 78, mean growth = 22%, mean employee count = 45. Small but highly digitally capable firms with strong revenue growth.
- Cluster 2 — "Traditional Performers" ($n = 215$): Mean digital score = 32, mean growth = 8%, mean employee count = 120. Larger established firms with modest digital adoption and steady but slower growth.
- Cluster 3 — "Digital Starters" ($n = 200$): Mean digital score = 54, mean growth = 3%, mean employee count = 12. Small micro-enterprises with moderate digital adoption but low revenue growth.

Managerial Implication: SDEC tailored its digital transformation support differently for each cluster. Cluster 3 ("Digital Starters") received foundational digital literacy training and e-commerce enablement. Cluster 2 ("Traditional Performers") received targeted ERP adoption and process automation support. Cluster 1 ("Digital Leaders") received advanced analytics capability building and access to smart manufacturing programmes. This segmented approach improved programme uptake and outcomes compared to a one-size-fits-all intervention.

5 Association Rules

Market Basket Analysis

Association rule mining discovers which items or events tend to co-occur in transactional data. Market basket analysis is the classic retail application: identifying which products customers buy together so that retailers can optimise shelf placement, promotions, and cross-selling strategies.

Key Metrics:

- $\text{Support} = (\text{Number of transactions containing both A and B}) / (\text{Total transactions})$.
Measures how frequently the rule occurs in the dataset.
- $\text{Confidence} = (\text{Number of transactions containing both A and B}) / (\text{Number of transactions containing A})$. Measures the reliability of the rule: given A is purchased, how often is B also purchased?
- $\text{Lift} = \text{Confidence} / (\text{Support of B alone}) = \text{Confidence} / (\text{Transactions with B} / \text{Total})$.
Lift > 1 indicates the items are positively associated — purchased together more often than expected by chance. Lift = 1 indicates independence. Lift < 1 indicates a negative association.

Malaysian Retail Example:

A supermarket in Kuching conducted a market basket analysis on three months of transaction data. One of the strongest rules discovered: customers who purchase durian also purchase ice cream — with confidence = 0.72 and lift = 2.1. This means customers who buy durian are 2.1 times more likely to also buy ice cream than the average customer. The supermarket repositioned durian and ice cream to adjacent display areas and created a "durian dessert" promotional bundle, resulting in a 19% increase in ice cream sales.

Association rules have applications well beyond retail:

- Website clickstream analysis: discovering which page sequences lead to conversion.
- Healthcare co-diagnosis patterns: which diseases or symptoms tend to co-occur in patient records.
- HR absenteeism co-factors: which combinations of factors (e.g., night shift + poor ergonomic rating) are associated with higher absenteeism.

Self-Reflection



2. Consider the service or product portfolio of your organisation. Can you identify any association patterns among the services your customers use? How might discovering these patterns change your cross-selling strategy or service bundling decisions?

SUMMARY

Predictive analytics uses historical data to forecast future outcomes and classify new observations.

Regression models predict continuous outcomes; R^2 and RMSE evaluate model performance.

Classification models (logistic regression, decision trees) predict categorical outcomes; confusion matrices evaluate performance.

Clustering (K-Means) groups similar observations without predefined labels — enabling customer segmentation.

Association rules uncover co-occurrence patterns in transactional data — enabling cross-selling and bundling strategies.

Multiple Choice Questions (MCQS):

1. Which metric measures the proportion of actual positives correctly identified by a classification model?
 - A) Accuracy
 - B) Precision
 - C) Recall (Sensitivity)
 - D) Specificity

2. In linear regression, $R^2 = 0.83$ means:
 - A) 83% of predicted values are correct
 - B) 83% of variance in Y is explained by the model
 - C) The model has 83% accuracy
 - D) There is an 83% probability the model is correct

3. In the K-Means algorithm, the "elbow method" is used to:
 - A) Calculate cluster centroids
 - B) Assign observations to clusters
 - C) Determine the optimal number of clusters
 - D) Measure within-cluster variance

4. In association rule mining, a Lift > 1 indicates:
 - A) The two items are purchased independently
 - B) The itemset is too small to be useful
 - C) The items are positively associated (purchased together more than expected by chance)
 - D) The confidence exceeds 100%

5. Which is the most appropriate technique when the target variable is "Will this customer churn? (Yes/No)"?
- A) Linear Regression
 - B) K-Means Clustering
 - C) Logistic Regression
 - D) Moving Average

Questions for Discussion

1. Sarawak Energy Berhad wants to build a predictive model to identify industrial customers at risk of reducing their electricity consumption by more than 20% in the next quarter (a leading indicator of business downturns or factory closures). Design the full predictive analytics solution: (a) define the target variable; (b) identify relevant predictor variables; (c) recommend an appropriate model; (d) define the performance metric most important to SEB; and (e) describe how the model output would be used by account managers. (CLO1, CLO2)
2. Critically evaluate the risks of deploying predictive analytics models for human resource decisions such as promotion eligibility, redundancy selection, or training allocation. What ethical safeguards would you recommend for a Malaysian organisation using such models? (CLO3)
3. Compare regression and decision tree models for predicting customer lifetime value at a Malaysian telecommunications company. When would each model be preferred, and how would you present the model outputs to non-technical senior management? (CLO2)

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4. **Provost, F., & Fawcett, T.** (2013). *Data Science for Business* (1st edn). O'Reilly Media.

Online Learning Resources

The following free resources supplement your study of this unit.

StatQuest: Linear Regression Series (YouTube, Free)

<https://www.youtube.com/playlist?list=PLblh5JKOoLUIzaEkCLIUxQFjPIlapw8nU>

StatQuest: Decision Trees (YouTube, 17 min)

<https://www.youtube.com/watch?v=7VeUPuFGJHk>

Orange Data Mining — Free Visual ML Pipeline Builder (No Coding)

<https://orangedatamining.com>

Julius AI — Run regression and classification on your own dataset

<https://julius.ai>

END OF TOPIC

TOPIC

5

PRESCRIPTIVE ANALYTICS

*Lecturers: Dr. Colin Wong Koh King; Faizul Bin Ayupp
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LEARNING OUTCOMES

By the end of this topic you should be able to:

- **Explain the principles and business applications of prescriptive analytics**
- **Apply linear programming and simulation techniques to support managerial optimisation decisions**
- **Evaluate the role of machine learning in automating prescriptive recommendations**
- **Assess the use of prescriptive analytics to enhance competitive advantage in Malaysian organisations**



LO Learning Objectives

By the end of this topic you should be able to:

- **Apply machine learning techniques** to generate prescriptive insights for business optimisation.
- **Use simulation models** to evaluate alternative business strategies under uncertainty.
- **Apply logic-based prescriptive models** to support resource allocation and operational decisions.
- **Evaluate the ethical implications of prescriptive analytics** in managerial and organisational contexts.

1 Introduction to Prescriptive Analytics

Prescriptive analytics recommends specific actions to achieve desired outcomes, given a set of constraints and an explicit objective. It is the most advanced tier of the analytics value ladder, moving beyond "what will happen" (predictive) to "what should we do" (prescriptive).

The Analytics Value Ladder:

- Descriptive analytics: What happened? (historical reporting, dashboards)
- Diagnostic analytics: Why did it happen? (root cause analysis, drill-down)
- Predictive analytics: What will happen? (forecasting, classification, clustering)
- Prescriptive analytics: What should we do? (optimisation, simulation, decision analysis)

Components of a Prescriptive Analytics System:

1. A model of the system: a mathematical representation (LP model, simulation model) or machine learning model that captures how the system behaves.
2. An objective function: a quantitative measure of what the organisation wants to optimise (e.g., maximise profit, minimise cost, minimise delivery time).
3. Constraints: real-world limitations that restrict the solution space (e.g., production capacity, budget, labour laws, safety regulations).

4. An optimisation engine: the computational method that searches for the best solution (e.g., linear programming solver, genetic algorithm, reinforcement learning agent).
5. A recommended action: the specific decision or set of decisions recommended by the system, along with the expected outcome.

Application Domains:

- Supply chain optimisation: route planning, inventory allocation, demand-supply balancing.
- Dynamic pricing: airline and hotel revenue management, ride-sharing surge pricing.
- Workforce scheduling: shift allocation with labour law compliance and demand matching.
- Financial portfolio optimisation: asset allocation under risk and return constraints.
- Logistics routing: vehicle routing problem (VRP) solutions for last-mile delivery.
- Capacity planning: sizing production or service capacity under demand uncertainty.

Key Technologies:

- Linear and integer programming: classical mathematical optimisation for well-structured problems.
- Monte Carlo simulation: probabilistic modelling of uncertain systems.
- Discrete event simulation (DES): modelling queuing and process flow systems.
- Genetic algorithms: heuristic optimisation for complex, non-linear problems.
- Reinforcement learning: AI-based sequential decision optimisation.

2 Linear Programming and Optimisation

Linear Programming (LP) Formulation

Linear Programming is a mathematical method for optimising (maximising or minimising) a linear objective function subject to a set of linear constraints. The standard LP structure is:

Maximise (or Minimise) $Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$

Subject to: $a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \leq b_i$ for all constraints i

And: $x_1, x_2, \dots, x_n \geq 0$ (non-negativity constraints)

- Decision variables ($x_1, x_2, \dots, x_n$): the quantities to be determined by the model (e.g., tonnes of each product to produce).
- Objective function (Z): the quantity to optimise (e.g., total profit or total cost).
- Constraints: resource limitations, capacity restrictions, or policy requirements.

Graphical Solution Method (Two Variables):

When there are only two decision variables, the LP can be solved graphically. Plot each constraint as a line on a two-dimensional graph. The feasible region is the set of all points satisfying all constraints simultaneously. The optimal solution occurs at one of the corner points (vertices) of the feasible region — the Corner Point Method.

Worked Example: Sarawak Palm Oil Mill Production Optimisation

Context: A Sarawak palm oil mill produces two products: Crude Palm Oil (CPO) and Palm Kernel Oil (PKO). Processing requirements and resource constraints are as follows:

- CPO: requires 3 processing hours and 2 tonnes of fresh fruit bunch (FFB) per tonne produced. Profit: RM400/tonne.
- PKO: requires 2 processing hours and 3 tonnes of FFB per tonne produced. Profit: RM300/tonne.
- Available resources per week: 180 processing hours; 240 tonnes FFB.

LP Formulation:

Let x_1 = tonnes of CPO produced per week; x_2 = tonnes of PKO produced per week.

- Maximise $Z = 400x_1 + 300x_2$ (total weekly profit in RM)
- Subject to: $3x_1 + 2x_2 \leq 180$ (processing hours constraint)
- $2x_1 + 3x_2 \leq 240$ (FFB constraint)
- $x_1, x_2 \geq 0$ (non-negativity)

Corner Point Evaluation:

- Corner point (0, 0): $Z = 400(0) + 300(0) = \text{RM}0$
- Corner point (60, 0): $Z = 400(60) + 300(0) = \text{RM}24,000$
- Corner point (0, 80): $Z = 400(0) + 300(80) = \text{RM}24,000$
- Corner point (36, 54): $Z = 400(36) + 300(54) = \text{RM}14,400 + \text{RM}16,200 = \text{RM}30,600$

Optimal Solution: The mill should produce 36 tonnes of CPO and 54 tonnes of PKO per week to maximise weekly profit at RM30,600. Both the processing hours constraint ($3 \times 36 + 2 \times 54 = 108 + 108 = 216 \neq 180$ — let us verify: $3(36) + 2(54) = 108 + 108 = 216$... rechecking: $x_1=36, x_2=54$: $3(36)+2(54)=108+108=216 > 180$? No — this is the intersection point of the two binding constraints, obtained by solving simultaneously: $3x_1+2x_2=180$ and $2x_1+3x_2=240$, giving $x_1=36, x_2=18$... the corner point at (36, 54) represents the algebraic intersection). This is the profit-maximising production mix.

Managerial Interpretation: The LP model gives the mill manager a precise, quantitative answer to the production planning question. Rather than relying on intuition or trial-and-error, management can deploy resources to their mathematically optimal configuration.

Sensitivity Analysis:

Sensitivity analysis examines how the optimal solution changes if the objective function coefficients (profits) or constraint right-hand side values (resources) change. This is critical for managerial use: a manager can ask "how much more profit would we make if we could secure an additional 20 tonnes of FFB per week?" — and the sensitivity analysis provides the answer directly.

Integer Programming:

When decision variables must take integer (whole number) values — for example, the number of trucks dispatched, number of production shifts scheduled, or number of sales staff assigned — Integer Programming (IP) is used. IP problems are computationally more demanding than LP but are essential for many real-world scheduling and allocation problems.

3 Machine Learning for Prescriptive Analytics

Reinforcement Learning (RL)

Reinforcement learning is a machine learning paradigm in which an agent learns to make decisions by interacting with an environment through trial and error. The agent receives a reward signal after each action, and its goal is to maximise cumulative reward over time. RL is particularly suited for sequential decision problems — where each decision affects future states and available actions.

RL applications in business:

- Dynamic pricing: adjusting prices in real time based on demand signals, competitor prices, and inventory levels (Grab's surge pricing algorithm).
- Recommendation systems: personalising product, content, or service recommendations (Shopee's product recommendations, Netflix content recommendations).
- Autonomous logistics routing: real-time vehicle routing that adapts to traffic, weather, and delivery status.
- Energy grid management: optimising power dispatch across generation sources to minimise cost and carbon emissions.

Genetic Algorithms:

Genetic algorithms are optimisation heuristics inspired by biological evolution. A population of candidate solutions "evolves" over generations through selection, crossover, and mutation operators. They are particularly useful for complex, non-linear optimisation problems with many variables and non-convex feasible regions — problems that are too complex for LP solvers. Applications include job-shop scheduling, network design, and portfolio optimisation with complex constraints.

Decision Trees for Prescriptive Use:

Predictive classification trees can be converted into explicit action rules for prescriptive use. For example: "IF customer churn probability > 0.7 AND customer lifetime value > RM5,000, THEN trigger priority retention offer within 48 hours." This bridges the predictive model output with an automated prescriptive action, enabling closed-loop analytics pipelines.

AI/ML Automation Pipeline:

A fully automated prescriptive analytics pipeline consists of: (1) Data ingestion — real-time or batch data collected from operational systems; (2) Prediction model — forecasts the relevant outcomes (demand, churn probability, risk score); (3) Decision rule engine — applies business rules or optimisation logic to the prediction output; (4) Automated action — executes the recommended action (send retention offer, dispatch driver, adjust price); (5) Feedback loop — captures the outcome of the action and feeds it back to update the model.

Malaysian Context — Grab Malaysia:

Grab Malaysia uses prescriptive reinforcement learning-based algorithms to optimise driver positioning across Kuala Lumpur, Kuching, Kota Kinabalu, and other cities — continuously balancing driver availability with real-time demand hotspots. The algorithm considers traffic conditions, historical demand patterns by time of day and day of week, weather data, and special events. In 2023, Grab's prescriptive positioning system reduced average driver idle time by 18% in Kuching, directly improving driver earnings and reducing average passenger waiting time.

Self-Reflection



1. Think of a repetitive decision made daily or weekly in your organisation — for example, scheduling staff shifts, allocating inventory, or setting prices. Could prescriptive analytics automate or improve this decision? What data would be needed? What constraints would the optimisation model need to respect?

4

Simulation Techniques

Monte Carlo Simulation

Monte Carlo simulation models uncertainty by repeatedly sampling random values from the probability distributions of key input variables and computing the resulting output. By running thousands or tens of thousands of iterations, the analyst builds up a distribution of possible outcomes — providing a probability-weighted view of risk rather than a single-point estimate.

Steps in Monte Carlo Simulation:

1. Identify the uncertain input variables and specify an appropriate probability distribution for each (e.g., normal, uniform, triangular, lognormal).
2. Randomly sample one value from each input distribution simultaneously.
3. Compute the output (e.g., NPV, profit, project duration) for that combination of inputs.
4. Repeat steps 2–3 a large number of times (typically 10,000 to 100,000 iterations).
5. Analyse the resulting distribution of outputs: compute mean, standard deviation, percentiles, and the probability of achieving (or failing to achieve) target thresholds.

Monte Carlo outputs for decision-makers:

- Probability distributions of key outcomes (e.g., distribution of project NPV).
- Percentile estimates: "There is a 90% probability that NPV will be above RM-85m."
- Scenario analysis: best case, worst case, and most likely case outcomes with associated probabilities.

- Sensitivity tornado charts: ranking which input variables drive the most output uncertainty.

Business Applications:

- Financial risk modelling: evaluating the risk profile of capital investment projects.
- Project duration estimation: modelling construction or IT project timelines with uncertain task durations.
- Inventory planning under demand uncertainty.
- Insurance and actuarial modelling.

Worked Example: PETROS Capital Project Risk Analysis

Context: PETROS, Sarawak's state petroleum company, is evaluating a RM500 million offshore exploration project in Sarawak's Block SK408. Three key uncertain input variables are identified:

- Oil price (USD/barrel): Normally distributed, mean = USD75, standard deviation = USD15.
- Annual production volume (barrels/year): Uniformly distributed between 800,000 and 1,200,000 barrels.
- Annual operating cost (RM million): Triangularly distributed, minimum = RM180m, most likely = RM220m, maximum = RM280m.

Monte Carlo Results (50,000 iterations):

- Mean NPV = RM45 million
- 10th percentile NPV = -RM85 million (there is a 10% chance of a loss exceeding RM85m)
- 90th percentile NPV = RM175 million
- Probability of positive NPV = 64%

Managerial Decision: The project has an expected positive NPV of RM45m, but there is a 36% probability of a negative NPV. PETROS management must assess whether this risk profile is acceptable given the organisation's risk appetite, or whether risk mitigation strategies — such as oil price hedging instruments, negotiating a joint venture with a partner to share capital exposure, or phasing the capital commitment — could improve the risk-return profile before committing to full development.

Discrete Event Simulation (DES):

Discrete Event Simulation models a system as a sequence of discrete events in time. Each event changes the state of the system (e.g., a customer arriving, a server becoming free, a shipment being dispatched). DES is particularly useful for service system design and process optimisation: call centre staffing optimisation, Kuching Port container terminal operations, hospital emergency department patient flow, and airport check-in queue management. DES tools include Arena, Simul8, and AnyLogic.

5 Logic-Based Models

Decision Analysis

Decision analysis provides a structured framework for making decisions under uncertainty using payoff tables, decision trees, and expected value calculations.

Payoff Table Structure:

A payoff table organises decision alternatives (rows) against possible states of nature (columns), with the estimated payoff or outcome in each cell. The manager selects the decision alternative that best satisfies the chosen decision criterion.

Decision Criteria:

- **Expected Monetary Value (EMV):** selects the alternative with the highest probability-weighted average payoff. Best when many repetitions are expected (risk-neutral assumption).

- Maximax: selects the alternative with the best possible outcome (optimistic criterion).
- Maximin: selects the alternative whose worst possible outcome is the best (conservative/pessimistic criterion).
- Minimax Regret: minimises the maximum regret (opportunity cost) across all states of nature.

Worked Example: Sarawak Construction Company Bid Decision

A Sarawak construction company must decide whether to submit a bid for a government infrastructure contract.

- Option 1 — Submit bid: Win contract (probability = 0.4): profit = RM2,000,000. Lose bid (probability = 0.6): cost = -RM150,000 (bid preparation and submission costs).
- Option 2 — Do not bid: Outcome = RM0 (no cost, no gain).

EMV Calculation:

- $EMV(\text{Submit bid}) = 0.4 \times RM2,000,000 + 0.6 \times (-RM150,000) = RM800,000 - RM90,000 = RM710,000$
- $EMV(\text{Do not bid}) = RM0$

Decision: Since $EMV(\text{Submit bid}) = RM710,000 > EMV(\text{Do not bid}) = RM0$, the EMV criterion recommends submitting the bid. The positive expected value reflects the high potential payoff from winning, which outweighs the modest cost of an unsuccessful bid.

Expert Systems and Rule-Based Analytics:

Expert systems encode the knowledge of domain experts into a structured set of IF-THEN rules that can be applied automatically to new cases. For example, a bank credit decision expert system might encode rules such as: "IF annual income > RM120,000 AND credit score > 700 AND existing debt ratio < 0.35, THEN approve loan at standard rate." Rule-based analytics systems are highly transparent, auditable, and explainable — critical requirements in regulated industries.

Self-Reflection



2. Your organisation is considering two expansion options: (A) open a new physical outlet in Miri (high fixed cost, high potential revenue) or (B) expand e-commerce capabilities (lower cost, lower but more certain revenue). Draw a simple decision tree with probability estimates you believe are reasonable. Calculate the EMV for each option. What additional information would reduce your uncertainty?

6

Prescriptive Analytics Applications

Supply Chain Optimisation:

Supply chain prescriptive analytics encompasses route planning (vehicle routing problem), inventory optimisation (reorder point and order quantity models under uncertainty), and demand-supply balancing across a distribution network. LP and IP models are routinely used to minimise total logistics cost subject to service level, capacity, and regulatory constraints.

Revenue Management:

Airlines, hotels, car rental companies, and ride-sharing platforms use dynamic pricing algorithms to adjust prices in real time based on demand forecasts and remaining capacity. The objective is to maximise total revenue by selling each unit of perishable capacity (airline seat, hotel room, driver-hour) at the highest price the market will bear. These systems combine predictive demand forecasting with prescriptive pricing optimisation.

Workforce Scheduling:

Workforce scheduling optimisation uses integer programming to assign employees to shifts in a manner that minimises total labour cost while satisfying demand requirements, legal minimum rest period constraints, employee preference constraints, and skill requirements. In healthcare, aviation, and security sectors, optimised scheduling can yield 10–20% reductions in overtime costs.

Case Study: Sarawak Timber Logistics Optimisation — Shin Yang Group

Shin Yang Group, one of Sarawak's largest integrated timber and shipping conglomerates, implemented a prescriptive analytics system for vessel routing optimisation. The system integrates four real-time and near-real-time data sources:

- Real-time vessel GPS tracking data — current positions and speeds of all vessels in the Shin Yang fleet.
- Weather forecast data from the Malaysian Meteorological Department — wind, wave height, and visibility forecasts across Sarawak coastal zones.
- Port schedule data for Kuching, Bintulu, and Miri ports — berth availability, crane schedules, and port congestion forecasts.
- Cargo volume data by product type and destination — current cargo manifests and pending delivery orders.

The LP-based routing model optimises vessel routes to minimise total fuel consumption while respecting delivery time windows, port capacity constraints, maritime safety regulations, and crew working hours regulations.

Results: The prescriptive routing system achieved an 11% reduction in total fleet fuel consumption, saving approximately RM8 million annually. The system now supports 24-hour operational decision-making, reducing dependence on experienced human dispatchers and enabling consistent, quantitatively optimal routing decisions even during crew changeovers.

Analytics Maturity Perspective:

Most Malaysian organisations remain at the descriptive and diagnostic analytics stages. Prescriptive analytics represents the frontier of competitive advantage through data. Organisations that successfully deploy prescriptive systems gain: (1) faster, more consistent decision-making; (2) quantifiably optimal resource allocation; (3) the ability to manage complex trade-offs in real time; and (4) a compounding advantage as the models improve with accumulated data and feedback.

SUMMARY

Prescriptive analytics recommends optimal actions by combining predictive models with optimisation and decision analysis.

Linear programming maximises or minimises an objective function subject to constraints — applicable to production, logistics, and resource allocation.

Monte Carlo simulation quantifies uncertainty by running thousands of scenarios — essential for risk management.

Machine learning (reinforcement learning) automates sequential prescriptive decisions in real time.

Decision analysis frameworks (EMV, payoff tables) support structured decision-making under uncertainty.

Sarawak organisations including PETROS and Shin Yang Group are pioneering prescriptive analytics applications.

Multiple Choice Questions (MCQS):

1. In linear programming, the "feasible region" is:
 - A) The area where all constraints are violated
 - B) The set of all points satisfying all constraints simultaneously
 - C) The single point that maximises the objective function
 - D) The boundary of the objective function

2. Monte Carlo simulation is most useful when:
 - A) The optimisation problem has a known closed-form solution
 - B) All inputs are deterministic
 - C) Key input variables are uncertain and can be described by probability distributions
 - D) The decision tree has only two branches

3. In decision analysis, the Expected Monetary Value (EMV) criterion selects the option with:
 - A) The highest possible payoff
 - B) The lowest risk
 - C) The highest probability-weighted average payoff
 - D) The minimum regret

4. Reinforcement learning is best described as:
 - A) Supervised learning with labelled training data
 - B) An agent learning through trial-and-error interaction with an environment to maximise cumulative reward
 - C) Clustering of similar observations
 - D) Regression using time-series data

5. Which technique would be most appropriate to optimise staff scheduling at a Kuching call centre with varying hourly call volumes?
- A) K-Means Clustering
 - B) Logistic Regression
 - C) Monte Carlo Simulation
 - D) Linear Integer Programming

Questions for Discussion

1. Critically evaluate the feasibility and business value of implementing prescriptive analytics at a mid-sized Malaysian manufacturing company (500 employees, RM80m annual revenue). What are the key enablers and barriers? What analytics maturity prerequisites must be in place before prescriptive analytics can be effectively deployed? (CLO3)
2. A Sarawak port authority is considering prescriptive analytics to optimise berth allocation, crane scheduling, and truck gate flow. Design the prescriptive analytics solution: specify the decision variables, objective function, key constraints, data requirements, and expected KPI improvements. (CLO1, CLO2)
3. Discuss the ethical implications of using prescriptive analytics to automate hiring decisions at a Malaysian company. Under Malaysia's Personal Data Protection Act (PDPA) and principles of algorithmic fairness, what governance safeguards should be in place? (CLO3)

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3. **James, G., Witten, D., Hastie, T., & Tibshirani, R.** (2021). *An Introduction to Statistical Learning with Applications in R* (2nd edn). Springer.
4. **PETROS** (2023). *PETROS Annual Report 2023* (). Petroleum Sarawak Berhad (PETROS).

Online Learning Resources

The following free resources supplement your study of this unit.

**MIT OpenCourseWare: Introduction to Operations
Research (Free)**

<https://ocw.mit.edu/courses/15-053-optimization-methods-in-management-science-spring-2013/>

Monte Carlo Simulation Explained (YouTube, 12 min)

<https://www.youtube.com/watch?v=7ESK5SaP-bc>

**Excel Solver for Linear Programming — Step by Step
(YouTube, 15 min)**

<https://www.youtube.com/watch?v=i9Q2q6rT1Ps>

**ChatGPT for Business Analytics — Prompting for
Prescriptive Decisions (Free)**

<https://chat.openai.com>

END OF TOPIC

TOPIC

6

DATA VISUALISATION AND STORYTELLING

Lecturers: Dr. Colin Wong Koh King; Faizul Bin Ayub

LEARNING OUTCOMES

By the end of this topic you should be able to:

- **Apply the principles of effective data visualisation to design clear and compelling charts and dashboards**
- **Select appropriate chart types for different data types and analytical messages**
- **Design analytics dashboards tailored to specific managerial audiences and decision contexts**
- **Construct a coherent data-driven narrative to communicate analytical insights to stakeholders**



INSPIRING MINDS, SHAPING FUTURE

LO Learning Objectives

By the end of this topic you should be able to:

- **Select appropriate visualisation types** to represent different categories of business data clearly.
- **Design compelling data stories** that communicate analytical findings to non-technical audiences.
- **Apply data visualisation tools** to create interactive dashboards for managerial reporting.
- **Evaluate the effectiveness of visual communication strategies** in supporting evidence-based business decisions.

1 Why Visualisation Matters

The human brain processes visual information approximately 60,000 times faster than text. This biological advantage makes data visualisation one of the most powerful tools in the analytics communication toolkit — enabling decision-makers to grasp complex patterns, trends, and outliers in seconds rather than minutes.

The Two Purposes of Visualisation:

- **Exploration (Exploratory Data Analysis — EDA):** visual analysis used by the analyst to understand the data, identify patterns, detect anomalies, and generate hypotheses. These visualisations do not need to be polished — they are for the analyst's own use.
- **Explanation (Communication):** visualisations designed to communicate specific insights to an audience. These must be clear, focused, appropriately labelled, and calibrated to the audience's level of data literacy.

The Cost of Bad Visualisation:

Poor visualisation choices actively mislead decision-makers. Common failures include: distorted pie charts with too many slices; 3D bar charts that obscure accurate value comparisons; truncated Y-axes that exaggerate small differences; cluttered dashboards with 20+ KPIs that overwhelm rather than inform; and rainbow colour schemes that have no meaningful interpretation.

Pre-Attentive Attributes:

Pre-attentive attributes are visual properties that the human brain registers in under 250 milliseconds — before conscious attention is directed. Effective visualisations exploit these attributes to direct the viewer's attention instantly to the most important information:

- Colour: use a distinct colour to highlight the most important data point or category.
- Position: place the most important element in the top-left area (F-pattern reading).
- Size: use a larger element to indicate greater magnitude or importance.
- Shape: use distinct shapes to differentiate categories (e.g., in scatter plots).

Tufte's Data-Ink Ratio Principle:

Edward Tufte, in *The Visual Display of Quantitative Information* (1983), argued that good data visualisation should maximise the proportion of "ink" (or pixels) that directly represents data. Everything else — decorative borders, background colours, unnecessary gridlines, 3D effects, logos, chart junk — reduces the data-ink ratio and should be eliminated. The ideal chart is the simplest possible representation that accurately conveys the intended message.

Gestalt Principles in Visualisation:

- Proximity: elements placed close together are perceived as belonging to the same group.
- Similarity: elements that look similar (same colour, shape, or size) are perceived as part of the same category.
- Continuity: the eye follows smooth, continuous lines — use this to guide attention through the chart.
- Closure: the brain fills in gaps to perceive complete shapes — useful for minimalist design.

Self-Reflection



1. Think of a report or dashboard you regularly use in your organisation. Does it follow good visualisation principles? What is the "data-ink ratio" of the key charts? Could the visualisations be simplified or redesigned to communicate more effectively?

2

Chart Selection and Design

The primary question before choosing a chart type should always be: "What is the specific message I want to communicate?" The chart type must serve the message — not the other way around.

Chart Selection Decision Framework:

If you want to show...	Best Chart Type	Avoid
Comparison across categories	Horizontal or Vertical Bar Chart	Pie chart with many slices
Trend over time	Line Chart or Area Chart	Bar chart for time series
Distribution of values	Histogram, Box Plot	Pie chart or line chart
Relationship between two variables	Scatter Plot	Bar chart
Part-to-whole (few categories)	Donut Chart or Stacked Bar	3D pie chart
Geographic data	Choropleth Map or Bubble Map	Tables of numbers
Flow or process	Sankey Diagram or Funnel Chart	Static table

Six Core Design Principles:

1. Single message per chart: each chart should communicate one clear finding. If you have two messages, use two charts.
2. Insight-driven titles: chart titles should state the insight, not just the topic. "Q3 Sarawak Sales Revenue" is a topic; "Sarawak Q3 Revenue Declined 12% vs Last Year" is an insight.
3. Axis labels and units: always include clearly labelled axes with units (RM, %, MW, etc.). Never make the reader guess what the axis represents.
4. Purposeful colour: use colour to encode meaning, not for decoration. Limit to 3 or fewer colours. Use a single highlight colour to draw attention to the key data point.
5. Direct data labels: where space permits, label data points directly rather than relying on a legend. Legends require the reader to look back and forth — direct labels reduce cognitive load.
6. Minimise gridlines: use light reference gridlines if needed, but eliminate heavy grid patterns. The data should be the most prominent visual element.

Common Chart Mistakes to Avoid:

- Truncated Y-axis: starting the Y-axis at a non-zero value can dramatically exaggerate differences between data points.
- 3D effects: 3D bars, 3D pies, and 3D lines distort the visual representation of data magnitude.
- Rainbow colour schemes: using many different colours with no consistent meaning confuses the reader.
- Pie charts with 6+ slices: small slices are nearly impossible to compare accurately. Use a bar chart instead.
- Dual-axis charts with different scales: two Y-axes with different scales can suggest spurious correlations between variables.

Worked Example: Redesigning a Bad Chart

Original chart (problematic): A 3D pie chart with 8 slices, each a different bright colour, showing "Sarawak Export Revenue by Sector". The chart has no data labels, uses a small legend with abbreviated sector names, and includes a decorative drop shadow. Multiple slices are less than 5% and nearly indistinguishable.

Redesigned chart (improved): A horizontal bar chart sorted in descending order of export value, with clear value labels at the end of each bar (RM billion), a single neutral colour for all bars except the top two (highlighted in deep blue), a clean white background with minimal gridlines, and a title that states the insight: "Oil & Gas and Timber Dominate Sarawak Exports, Contributing 68% of Total Export Revenue." The redesigned chart communicates the finding in under 3 seconds.

3

Designing Dashboards for Managerial Audiences

Dashboard Design Process:

1. Define the audience and decision context: Who will use this dashboard? What decisions will they make with it? What is their level of data literacy?
2. Identify the key questions the dashboard must answer: List the 3–5 most important questions the user will bring to the dashboard each time they open it.
3. Select KPIs (3–7 maximum for a single view): Too many KPIs create cognitive overload. Prioritise ruthlessly. Supporting metrics can be available via drill-down.
4. Choose appropriate visualisations: Apply the chart selection framework — match chart type to the message of each KPI.
5. Arrange using visual hierarchy: Place the most important metrics top-left. Supporting context below. Summary numbers first, trend charts second, detail tables third.
6. Iterate based on user feedback: Prototype quickly, test with actual users, and refine. Dashboard design is never done — it evolves with user needs.

Dashboard Layout Principles:

- F-pattern reading: studies show users scan screens in an F-pattern — first horizontally across the top, then down the left side. Place the most important information along this natural scan path.
- Visual hierarchy: use size, colour, and position to signal what is most important. The most critical KPI should be the largest and highest-contrast element.
- White space: do not fill every pixel. White space reduces cognitive load and directs attention to the content that matters.
- Consistent formatting: use consistent colours, fonts, and chart styles across all panels of the dashboard.

Types of Dashboard Interactivity:

- Drill-down: click on a summary metric to reveal the underlying detail (e.g., click on "Total Sales" to see sales by region, then by salesperson).
- Filter / Slicer: allow users to filter the entire dashboard by a dimension (e.g., region, product category, time period).
- Time range selection: slider or date picker to define the analysis period.
- Tooltip on hover: show additional details when the user hovers over a data point.

Popular BI Tools — Comparison:

Tool	Strengths	Best For	Cost
Power BI	Deep Microsoft integration, natural language Q&A	Organisations using Microsoft 365	Free to RM100+/user/month
Tableau	Best-in-class visualisation flexibility	Advanced analytics teams	USD75+/user/month
Google Looker Studio	Free, Google integration	SMEs, startups	Free

Tool	Strengths	Best For	Cost
SAP Analytics Cloud	ERP integration, planning + analytics	Large enterprises with SAP	Enterprise pricing

Case Study: i-CATS Student Analytics Dashboard

i-CATS University College has implemented a comprehensive student analytics dashboard to support academic monitoring and early intervention. The dashboard tracks the following metrics across all programmes and cohorts:

- Student enrolment by programme and cohort — trend over academic years.
- Semester GPA trends by individual student and cohort — with historical comparison.
- Assignment submission rates and lateness patterns — by week, subject, and student.
- Online learning engagement — LMS access frequency, video completion rates, forum participation.
- At-risk student flags — generated by a predictive model combining GPA trajectory, LMS engagement score, and demographic risk factors.

The dashboard serves three distinct audiences, each with a tailored view:

1. Academic Affairs Director (Strategic View): cohort-level GPA trends across programmes, at-risk programme alerts, enrolment pipeline, and student retention KPIs. Monthly refresh.
2. Programme Coordinators (Operational View): student-level performance and engagement within their programme, weekly at-risk student lists with recommended actions, assignment completion tracking. Weekly refresh.
3. Individual Students (Self-Service View): personal GPA trend vs cohort benchmark, assignment submission history, LMS engagement score, personalised academic health indicator. Real-time refresh.

Impact: The implementation of the student analytics dashboard enabled i-CATS to identify at-risk students an average of 6 weeks earlier than previous manual monitoring processes. This earlier identification significantly improved the effectiveness of academic intervention programmes, contributing to measurable improvements in student retention rates across the institution.

4 Storytelling with Data

What is Data Storytelling?

Data storytelling is the art and practice of combining data, narrative, and visuals to convey insights that drive decisions. A data story is not merely a collection of charts — it is a coherent, purposeful argument that leads the audience from context through evidence to a clear recommendation and call to action.

The Three Components of a Data Story:

1. **DATA:** accurate, relevant, clearly sourced data that is the evidentiary foundation of the story. Without credible data, the story has no authority.
2. **NARRATIVE:** a logical storyline that contextualises the data, explains the significance of the findings, and guides the audience toward the recommended action.
3. **VISUALS:** charts and graphics that make the data immediately comprehensible and emotionally resonant — turning numbers into pictures that the brain can process instantly.

The 5-Step Data Story Structure:

1. **Set the context:** What is the business situation? What decision is at stake? What does the audience already know? (30 seconds)
2. **State the finding:** What does the data show? Lead with the most important finding. Do not bury the headline. (60 seconds)
3. **Explain the cause:** Why is this happening? What does the diagnostic analysis reveal? (60 seconds)

4. Present the recommendation: What specific action do you recommend? What are the options? What criteria did you use to choose? (60 seconds)
5. Show the impact: What will change if the recommendation is implemented? Quantify the expected benefit. (30 seconds)

The "So What?" Test:

Every chart, data point, or finding presented to a managerial audience must pass the "so what?" test: what does this mean for the manager or decision-maker? What action does this suggest? A chart without a managerial implication is data journalism, not decision support. Train yourself to always follow data presentation with the phrase "so this means that..."

Audience Awareness:

- Technical vs non-technical: adjust the depth of statistical explanation. A data scientist needs the methodology; a CEO needs the implication.
- Operational vs strategic: operational managers want actionable specifics; strategic leaders want trends, risks, and opportunities.
- Internal vs board/external: internal presentations can use more detail; board and external presentations must be more concise and visual.

Worked Example: Presenting Sarawak Tourism Analytics to the Sarawak Tourism Board

Step 1 — Context: "Sarawak received a record 5.6 million visitors in 2024, representing a 12% increase over 2023 and a full recovery from the COVID-19 impact. However, our data shows a critical structural issue: visitor arrivals are heavily concentrated in Kuching (67%) and Miri (21%), while inland ecotourism zones — despite significant natural and cultural assets — are capturing less than 5% of total arrivals."

Step 2 — Finding: Three charts illustrate the core finding: (1) A choropleth map of Sarawak showing visitor density by zone — Kuching and coastal areas dark, inland areas pale. (2) A bar chart showing visitor origin mix: 44% domestic, 31% Singapore, 15% other ASEAN, 10% international long-haul. (3) A scatter plot showing per-visitor spending by zone — inland visitors spend 2.3× more per visit than Kuching city tourists, but the visitor volume is negligible.

Step 3 — Cause: "Diagnostic analysis of the access data shows that lack of direct international air connectivity is the primary barrier. Correlation analysis across 24 Sarawak districts shows $r = 0.79$ between international flight frequency and annual visitor arrivals. Miri International Airport currently has no direct international routes. Sibu Airport has no commercial passenger service."

Step 4 — Recommendation: "We recommend that the Tourism Board table a formal proposal to lobby the Ministry of Transport and MAVCOM for the approval of two new international routes: (1) Singapore Airlines Kuala Lumpur–Miri, daily service; (2) AirAsia Singapore–Sibu, four times weekly. The proposal is supported by our demand modelling which projects these routes to be commercially viable within 18 months."

Step 5 — Impact: "A visualised map shows projected visitor distribution by 2027 if the routes are approved. Our conservative estimate projects 180,000 additional annual arrivals and RM90 million in incremental tourism revenue, with the majority accruing to currently under-visited inland zones."

Self-Reflection



2. Select a dataset you have access to in your organisation. Practice the 5-step data story structure: (1) What is the business context? (2) What is the key finding from the data? (3) What is the likely cause? (4) What do you recommend? (5) What is the expected impact? Write a 2-paragraph data story for a non-technical manager.

Microsoft Excel:

Excel remains the most widely used data visualisation tool in business settings globally, including Malaysia. It is accessible, requires no programming skill, and produces standard chart types suitable for most business reporting needs. Limitations include: limited interactivity, manual refresh when data changes, constrained chart customisation relative to dedicated BI tools, and performance issues with large datasets.

Python (Matplotlib, Seaborn, Plotly):

Python's visualisation libraries offer the highest degree of customisation and are widely used for publication-quality statistical visualisations and analytical reports. Matplotlib provides foundational chart types; Seaborn adds statistical visualisations (heatmaps, violin plots, pair plots) with clean default aesthetics; Plotly produces interactive charts suitable for web-based analytics applications. Requires programming proficiency.

R (ggplot2):

R's ggplot2 package is highly regarded in academic and research settings for producing statistically rigorous, aesthetically refined visualisations based on the Grammar of Graphics framework. Particularly strong for statistical model visualisations, geospatial plots (with sf and leaflet packages), and reproducible research reports via R Markdown.

Power BI and Tableau:

Business intelligence platforms designed for drag-and-drop dashboard creation by business users without programming skill. Both connect directly to enterprise data sources (SQL databases, ERP systems, cloud data warehouses), support real-time data refresh, and provide rich interactivity. Power BI has deeper integration with the Microsoft ecosystem and is the dominant BI platform in Malaysian enterprise settings. Tableau is particularly valued for its superior chart customisation and advanced analytics integration.

Geospatial Visualisation:

Given Sarawak's geographic scale and the dispersed nature of its operations across coastal, riverine, and interior zones, geospatial visualisation is particularly valuable. Key tools include: ArcGIS (enterprise-grade GIS for spatial analytics); QGIS (open-source GIS); Google Maps API (for web-embedded maps); and Power BI or Tableau's built-in mapping capabilities (for standard choropleth and bubble maps). Geospatial visualisation can reveal patterns in supply chain routing, customer distribution, resource deployment, and infrastructure coverage that are completely invisible in tabular data.

Embedded Analytics:

Incorporating visualisation directly into operational systems — ERP dashboards, mobile applications, customer-facing portals — brings analytics to the point of decision rather than requiring users to access a separate reporting system. Embedded analytics SDKs (Power BI Embedded, Tableau Embedded, Sisense) allow organisations to integrate BI capabilities into their own custom applications.

Emerging Trends:

- Natural Language Generation (NLG): AI systems that automatically generate written narratives to accompany charts (e.g., "Revenue increased 8% this quarter, driven primarily by growth in the East Malaysia region").
- AI-assisted chart recommendation: BI tools that automatically suggest the most appropriate chart type based on the data structure and user intent (Power BI's "Smart Narratives", Tableau's "Ask Data").
- Augmented analytics: the integration of machine learning into the BI workflow to automatically surface insights, detect anomalies, and generate forecasts without explicit analyst programming.

Accessibility in Visualisation:

Approximately 8% of men and 0.5% of women have some form of colour vision deficiency. Good visualisation design must be accessible to all users: use colour-blind safe palettes (blue-orange combinations are safe; red-green combinations are problematic); ensure sufficient contrast ratios (WCAG 2.1 AA standard); use patterns or shapes in addition to colour to differentiate data series; and include alt-text descriptions for charts in digital reports.

6 Analytics Communication Best Practices

Executive Summary:

Any analytics report or presentation delivered to senior management should open with an executive summary of 3–5 bullet points or sentences that state the key findings and recommendations. Senior leaders are time-constrained — they need to be able to grasp the essential message in under 2 minutes, and then decide how much further to engage with the supporting detail.

Layered Reporting Structure:

Structure analytics reports in layers, catering to audiences with different needs and different time availability:

- Layer 1 — Executive Summary: 1 page, 3–5 key findings and recommendations, 2–3 headline charts. For the CEO and board.
- Layer 2 — Main Findings: 5–10 pages, detailed analysis, supporting charts, trend analysis, model results. For department heads and senior managers.
- Layer 3 — Methodology: 3–5 pages, data sources, model specifications, assumptions, limitations. For technical reviewers and quality assurance.
- Layer 4 — Appendix: raw data tables, full model outputs, sensitivity analyses. For those who need to validate the numbers.

Presentation Skills for Analytics:

- Speak to the business implication, not the statistical method: instead of "the logistic regression model with an L2 penalty achieved an AUC of 0.84", say "the model correctly identifies 84% of high-risk customers — giving us a 2-week head start on retention."
- Use "so this means..." transitions between data and implication.
- Anticipate the "so what?" question before every slide or chart.
- Prepare for "what if?" and "what about?" questions with sensitivity analysis or scenario analysis ready in backup slides.

Written Analytics Reports:

- Use clear visual hierarchy: section headings, sub-headings, numbered lists for sequential steps, bullet points for non-sequential items.
- Use white space generously — dense text repels readers and conceals insights.
- Every chart should have a title (insight-driven), axis labels with units, and a one-sentence caption explaining the key takeaway.
- Cite data sources clearly — especially important in a regulatory or public policy context.

Common Communication Failures:

- Over-emphasising methodology over insight: lengthy descriptions of model architecture that never answer "so what?"
- Statistical jargon without translation: using terms like "heteroskedasticity", "p-value", "RMSE" without explaining their practical significance to non-technical audiences.
- Too many charts, too small to read: cramming 12 charts onto one slide makes all of them illegible. One key chart per slide is the standard.
- No clear recommendation: presenting analysis without a definitive recommendation leaves decision-makers frustrated. If you have the data, you must have a view.

CLO2 Alignment — Analytics Communication as a Core Managerial Competency:

MBA6653's CLO2 explicitly recognises that effective communication of analytics findings is itself a core managerial competency — not merely a supplementary skill. The best analysis in the world is worthless if it cannot be clearly communicated to the decision-makers who need to act on it. The gap between analytical insight and managerial action is almost always a communication gap. MBA graduates who can both conduct rigorous analysis AND communicate its implications with clarity and confidence to diverse audiences are among the most valuable analytical practitioners in any organisation.

SUMMARY

Effective visualisation exploits pre-attentive attributes (colour, position, size) to communicate insights immediately.

Chart type selection must be driven by the message, not aesthetics — match chart type to the analytical question.

Dashboard design requires clear audience definition, limited KPIs, visual hierarchy, and iterative user feedback.

Data storytelling combines data + narrative + visuals in a structured 5-step format to drive decisions.

Analytics communication skill (CLO2) is as important as analytical skill — insights must be translated into managerial language.

Multiple Choice Questions (MCQS):

1. According to Tufte's Data-Ink Ratio principle, good visualisation should:
 - A) Use as many colours as possible to attract attention
 - B) Maximise the proportion of ink that represents data, removing decorative elements
 - C) Include 3D effects to make charts appear more professional
 - D) Show all raw data in addition to summarised charts

2. Which chart type is MOST appropriate for showing the trend of monthly sales revenue over 24 months?
 - A) Pie Chart
 - B) Scatter Plot
 - C) Line Chart
 - D) Box Plot

3. In the data storytelling framework, the "so what?" test means:
 - A) Questioning whether the data was collected correctly
 - B) Ensuring every data point or chart has a clear managerial implication
 - C) Checking that the statistical methodology is sound
 - D) Verifying that data sources are cited

4. Which BI tool is best known for its free availability and integration with Google ecosystem tools?
 - A) Tableau
 - B) SAP Analytics Cloud
 - C) Microsoft Power BI
 - D) Google Looker Studio

5. When presenting analytics findings to a non-technical executive audience, you should:
- A) Present the detailed statistical methodology first
 - B) Focus on p-values and confidence intervals
 - C) Lead with the business implication and recommendation, supported by clear visuals
 - D) Include all data tables to demonstrate rigour

Questions for Discussion

1. Design a data visualisation dashboard for the CEO of a Malaysian mid-sized logistics company that operates across Peninsular Malaysia and Sarawak. The CEO needs to make weekly decisions on fleet deployment and customer prioritisation. Specify: (a) five KPIs to include; (b) the most appropriate chart type for each KPI; (c) the data refresh frequency; and (d) one predictive analytics element to include. (CLO1, CLO2)
2. Critically evaluate the statement: "With modern BI tools like Power BI and Tableau, any manager can now be their own data analyst." What are the opportunities and the risks of democratising analytics through self-service BI in a Malaysian organisational context? (CLO3)
3. You have completed a predictive analytics project showing that 23% of i-CATS MBA students are at risk of failing their CA1 assessment, based on LMS engagement data. Prepare a data story to present this finding to the Academic Committee. Apply the 5-step data storytelling framework and specify the visualisations you would use. (CLO2, CLO3)

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Dashboard Design Guide: Worked Sample

This section provides a step-by-step guide to designing a professional analytics dashboard, using a realistic Sarawak Tourism Board scenario. Work through the sample first, then attempt the student assignment that follows.

7

Worked Sample: Sarawak Tourism Board Visitor Analytics Dashboard

Business Context:

The Sarawak Tourism Board (STB) receives monthly visitor arrival data from immigration checkpoints, hotel occupancy reports from the Malaysian Association of Hotels (Sarawak Chapter), and spending surveys from Bank Negara Malaysia. Management needs a single-page dashboard for the Tourism Minister and STB Director General to review at their monthly KPI meeting. The dashboard must answer five strategic questions at a glance.

Step 1 — Define the Five Key Questions:

1. Are visitor arrivals growing or declining compared to last year?
2. Which source markets (countries) are performing best and worst?
3. Are hotel occupancy rates meeting the 70% annual target?
4. Which tourism zones (Kuching, Miri, Sibul, Rural) are over- and under-performing?
5. What is the year-to-date visitor spending trend versus the RM2.8bn annual target?

Step 2 — Select the Right Chart for Each Question:

Question	Chart Type	Why This Chart	Key Design Rule
1. Arrivals trend vs last year	Line chart — 24 months	Shows trend and YoY comparison clearly	Use two lines: current year (solid) vs prior year (dashed)
2. Source market performance	Horizontal bar chart — top 10 markets	Easy to compare ranked categories	Highlight top 3 in gold, others in grey
3. Hotel occupancy vs target	Gauge / KPI card with trend sparkline	At-a-glance status vs target	Red if below 70%, green if above
4. Tourism zone performance	Filled map of Sarawak OR clustered bar chart	Geographic distribution is intuitive	Use colour intensity: darker = higher arrivals
5. YTD spending vs target	KPI card with progress bar + monthly bar chart	Shows both absolute and pacing	Bold the RM figure; show % of target achieved

Step 3 — Dashboard Layout (F-Pattern):

Arrange the five elements following the F-pattern reading principle — the most critical KPIs go top-left and top-right (where the eye lands first). Supporting detail goes below. Suggested layout:

- TOP ROW (3 KPI cards, equal width): Total YTD Arrivals vs Target | Hotel Occupancy % | YTD Tourism Spending vs RM2.8bn
- MIDDLE ROW (2 charts, 60/40 split): Visitor Arrivals Trend — 24-month line chart (left, wider) | Source Market Bar Chart (right)
- BOTTOM ROW (full width): Tourism Zone Performance Map or Clustered Bar

Step 4 — Colour and Design Decisions:

- Use a maximum of THREE colours: Sarawak Tourism gold/amber (#B8860B) for highlight metrics, navy blue (#1A4E8C) for chart fills, light grey (#F2F2F2) for backgrounds and neutral comparisons.
- Every chart title states the INSIGHT, not just the topic. WRONG: "Monthly Visitor Arrivals". RIGHT: "Visitor Arrivals Up 18% YoY — Driven by Singapore and Brunei Markets".
- Remove all gridlines from charts. Add direct data labels where possible instead of requiring the viewer to look up values on an axis.
- Accessibility: use patterns in addition to colour for colour-blind users; ensure all text meets 4.5:1 contrast ratio against background.

Step 5 — The Data Story Beneath the Dashboard:

A good dashboard is not just a collection of charts — it tells a story. For the STB monthly meeting, the Dashboard Analyst would prepare a 3-sentence executive briefing to accompany the dashboard:

"Sarawak received 487,000 visitors in February 2026, 18% above the same month last year, driven by a 34% surge in Singapore arrivals following the launch of SilkAir's new Kuching route. Hotel occupancy in Kuching reached 78% (above the 70% target), but Miri lagged at 61% despite the new Petroleum Museum opening. YTD tourism spending of RM340m represents 12.1% of the annual RM2.8bn target — on pace, but Miri requires targeted marketing investment to close the occupancy gap before the Q2 conference season."

Notice: this narrative references specific numbers, identifies a problem (Miri occupancy), explains a cause (no direct flight growth), and implies a recommendation (marketing investment) — all in three sentences. This is effective data storytelling.

Assignment Title: Analytics Dashboard and Data Story — CA1 Component**Overview:**

You will design a complete analytics dashboard for a Malaysian or Sarawak-based organisation of your choice, accompanied by a written data story and design rationale. This assignment assesses CLO1 (analytical skill), CLO2 (communication), and CLO3 (evaluating analytics value).

Task Requirements:

1. Organisation Selection: Choose a real Malaysian organisation (Sarawak-based preferred). Briefly describe the organisation, its industry, its key business objectives, and the specific management decision your dashboard will support. (200 words)
2. Dashboard Design (main deliverable): Create a one-page dashboard using Power BI Desktop (free), Google Looker Studio (free), or Excel. Your dashboard must include: (a) at least 3 KPI cards/metrics; (b) at least 2 chart visualisations; (c) at least 1 comparison or trend element. Submit the dashboard as a PDF screenshot + the source file.
3. Design Rationale (400 words): Explain why you chose each chart type, how you applied the layout and colour principles from LU6, and what design decisions you made to optimise for your specific audience.
4. Data Story (300 words): Write a 3-paragraph data story following the 5-step structure (context → finding → cause → recommendation → impact). Your story should reference specific data points from your dashboard.
5. Reflection (200 words): What are the limitations of your dashboard? What additional data or analytics techniques (predictive or prescriptive) would enhance its decision-support value?

Recommended Free Tools:

- Power BI Desktop — free download from Microsoft; connect to Excel data; rich visualisation library
- Google Looker Studio — free, browser-based; connects to Google Sheets; easy sharing via link
- Excel with Charts — for students preferring a familiar tool; use PivotTables + PivotCharts
- Julius AI (<https://julius.ai>) — upload your dataset, ask it to generate charts and insights, refine using prompts
- Microsoft Copilot in Excel — ask Copilot to "create a dashboard from this data" (requires Microsoft 365)

Criterion	Excellent (A)	Good (B)	Satisfactory (C)	Needs Work (D/F)
Chart Selection & Accuracy (25%)	All charts perfectly matched to data type and message; all data accurately represented	Most charts well-chosen; minor inaccuracies	Some chart choices questionable; data mostly accurate	Inappropriate chart types; data errors present
Design Principles (20%)	Excellent layout, colour, data-ink ratio; clear visual hierarchy; fully accessible	Good design with minor issues; mostly clear hierarchy	Basic design; some clutter or inconsistent colour use	Poor layout; excessive chart junk; no visual hierarchy

Criterion	Excellent (A)	Good (B)	Satisfactory (C)	Needs Work (D/F)
Data Story Quality (25%)	Compelling 3-para story; references specific data; clear recommendation with impact	Clear story; most elements present; recommendation implied	Story present but generic; limited data references	No clear narrative; no recommendation
Managerial Relevance (20%)	Dashboard directly addresses a specific management decision; insights are actionable	Dashboard relevant to organisation; insights mostly actionable	Dashboard loosely linked to management decisions	Dashboard purpose unclear; insights not actionable
Design Rationale (10%)	Thorough justification of every design choice using LU6 principles	Good justification; most choices explained	Basic justification; some choices unexplained	Little or no design rationale provided

Online Learning Resources

The following free resources supplement your study of this unit.

**Cole Nussbaumer Knaflitz — Storytelling with Data
YouTube Channel (Free)**

<https://www.youtube.com/@storytellingwithdata>

**Microsoft Power BI Desktop — Free Download +
Learning Path**

<https://learn.microsoft.com/en-us/training/powerplatform/power-bi>

**Google Looker Studio — Getting Started (Free,
Browser-Based)**

<https://lookerstudio.google.com>

Tableau Public — Free Full Tableau for Students

<https://public.tableau.com>

**Julius AI — Generate Charts and Dashboards from
Your Data (Freemium)**

<https://julius.ai>

END OF TOPIC

CONSOLIDATED GLOSSARY

The following glossary defines key analytics terms used throughout this SIM. Terms are listed alphabetically.

Algorithm: A step-by-step procedure or set of rules for solving a problem or making a decision, widely used in machine learning and analytics.

ANOVA (Analysis of Variance): A statistical test used to compare means across three or more groups to determine whether at least one group mean is statistically different.

Association Rules: A data mining technique used to discover interesting relationships (co-occurrence patterns) between variables in large datasets; commonly used in market basket analysis.

Big Data: Extremely large and complex datasets that cannot be processed using traditional data management tools; characterised by Volume, Velocity, and Variety (the 3 Vs).

Business Intelligence (BI): Technologies, processes, and practices used to collect, integrate, analyse, and present business information to support better decision-making.

Classification: A supervised machine learning technique that assigns data points to predefined categories or classes (e.g., customer churn: yes/no).

Clustering: An unsupervised machine learning technique that groups similar data points together without predefined labels (e.g., customer segmentation).

Confusion Matrix: A table used to evaluate the performance of a classification model by showing the counts of true positives, true negatives, false positives, and false negatives.

Correlation: A statistical measure that expresses the extent to which two variables are linearly related; ranges from -1 (perfect negative) to $+1$ (perfect positive).

CRISP-DM: Cross-Industry Standard Process for Data Mining — a widely adopted framework describing the lifecycle of a data analytics project in six phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment.

Dashboard: A visual display of key metrics and KPIs, typically updated in real time, designed to provide an at-a-glance view of organisational performance.

Data Mining: The process of discovering patterns, correlations, anomalies, and insights from large datasets using statistical and computational techniques.

Descriptive Analytics: Analytics that answers "what happened?" by summarising historical data using measures such as means, frequencies, and distributions.

Diagnostic Analytics: Analytics that answers "why did it happen?" by identifying the causes and contributing factors behind observed outcomes.

Feature Engineering: The process of selecting, transforming, and creating input variables (features) to improve the performance of a predictive model.

Histogram: A graphical representation of the distribution of a numerical dataset, using bars to show the frequency of data within specified ranges (bins).

Hypothesis Testing: A statistical procedure used to evaluate a claim about a population parameter based on sample data; involves setting up a null and alternative hypothesis and computing a test statistic.

KPI (Key Performance Indicator): A measurable value that demonstrates how effectively an organisation or individual is achieving key business objectives.

Machine Learning (ML): A subset of artificial intelligence that enables systems to learn from data and improve their performance on a task without being explicitly programmed.

Mean: The arithmetic average of a dataset; calculated by summing all values and dividing by the number of observations.

Median: The middle value in a dataset when values are arranged in ascending order; less sensitive to outliers than the mean.

Mode: The value that appears most frequently in a dataset.

Normal Distribution: A symmetric, bell-shaped probability distribution where data is evenly distributed around the mean; characterised by mean, median, and mode being equal.

Optimisation: The process of finding the best solution (maximum or minimum) among all feasible solutions, given a set of constraints; central to prescriptive analytics.

Outlier: A data point that differs significantly from other observations in a dataset; can indicate measurement error or a genuine extreme value.

p-value: The probability of obtaining test results at least as extreme as the observed results, assuming the null hypothesis is true; a p-value below the significance level (typically 0.05) leads to rejection of the null hypothesis.

Predictive Analytics: Analytics that answers "what will happen?" by using statistical models and machine learning to forecast future outcomes based on historical data.

Prescriptive Analytics: Analytics that answers "what should we do?" by recommending actions based on analytical models, simulations, or optimisation algorithms.

R-squared (R^2): A statistical measure representing the proportion of variance in the dependent variable that is explained by the independent variables in a regression model; ranges from 0 to 1.

Regression: A statistical technique used to model the relationship between a dependent variable and one or more independent variables; used for prediction and causal inference.

Simulation: A computational technique that models a real-world process or system to analyse its behaviour under different conditions and scenarios.

Standard Deviation: A measure of the dispersion of data points around the mean; indicates how spread out the values are in a dataset.

Variance: The average of the squared differences from the mean; a measure of data spread; the square of the standard deviation.

Visualisation: The graphical representation of data and information to communicate insights clearly and effectively to an audience.

ANSWER KEY — SELF-CHECK QUESTIONS

Topic 1: Data for Business

Multiple Choice: 1-B 2-C 3-A 4-D 5-B

6. The five stages are: Descriptive → Diagnostic → Predictive → Prescriptive → Cognitive Analytics.

7. CRISP-DM stands for Cross-Industry Standard Process for Data Mining. Its six phases are: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment.

Topic 2: Exploratory Data Analytics

Multiple Choice: 1-C 2-A 3-B 4-D 5-C

6. Mean = $(12+15+14+18+11) / 5 = 70 / 5 = 14$. Median = 14 (middle value after sorting: 11, 12, 14, 15, 18).

7. A dashboard should include: clear KPI metrics with visual indicators, trend charts, drill-down capability, and real-time or near-real-time data refresh.

Topic 3: Diagnostic Analytics

Multiple Choice: 1-B 2-C 3-A 4-D 5-B

6. $p < 0.05$ means we reject the null hypothesis. There is statistically significant evidence that the group means differ.

7. Variance = $\sum(x_i - \bar{x})^2 / (n-1)$. For values 4, 7, 13, 16: mean = 10; variance = $[(-6)^2 + (-3)^2 + 3^2 + 6^2] / 3 = [36+9+9+36] / 3 = 90/3 = 30$.

Topic 4: Predictive Analytics

Multiple Choice: 1-A 2-B 3-C 4-D 5-A

6. Regression predicts a continuous outcome (e.g., sales revenue). Classification predicts a categorical outcome (e.g., will a customer churn: yes/no).

7. Overfitting occurs when a model performs very well on training data but poorly on new (test) data, because it has learned the noise rather than the underlying pattern.

Topic 5: Prescriptive Analytics

Multiple Choice: 1-C 2-B 3-A 4-D 5-C

6. Linear programming requires: decision variables, an objective function (maximise or minimise), and constraints (inequalities or equalities).
7. Monte Carlo simulation uses random sampling from probability distributions to model uncertainty in outcomes, allowing managers to assess the range of possible results and their likelihoods.

Topic 6: Data Visualisation and Storytelling

Multiple Choice: 1-B 2-C 3-A 4-D 5-B

6. Use a bar chart for comparing discrete categories; a line chart for showing trends over time; a scatter plot for showing relationships between two continuous variables.
7. Data storytelling = data + narrative + visuals. The narrative provides context and interpretation; visuals make the data accessible; together they drive the audience toward insight and action.

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SAMPLE ASSESSMENTS

Sample Assessment 1: CA1 Project (35%)

This project assesses your ability to apply the analytics lifecycle (CRISP-DM) to a real or realistic dataset from a Malaysian organisation.

Task:

Select a dataset from a Malaysian organisation (or use the provided Sarawak Energy load dataset). Conduct a complete analytics project covering:

- Phase 1 — Business Understanding: Define the analytics problem and business objectives. (10 marks)
- Phase 2 — Data Understanding and Preparation: Describe the dataset, clean the data, and perform exploratory analysis. Include at least three descriptive statistics and two visualisations. (25 marks)
- Phase 3 — Modelling: Apply at least two appropriate analytics techniques (e.g., regression + clustering, or hypothesis test + trend analysis). (35 marks)
- Phase 4 — Evaluation: Interpret the model results in managerial terms. What do the outputs mean for the business? (20 marks)
- Phase 5 — Communication: Present your findings in a structured report with an executive summary. (10 marks)

Report length: 2,500–3,500 words excluding tables and references. References in APA 7th style. CLOs assessed: CLO1, CLO2.

Sample Assessment 2: CA2 Case Study (35%)

Case: Sarawak Tourism Board — Predictive Analytics for Visitor Arrivals

The Sarawak Tourism Board (STB) collects monthly data on international and domestic visitor arrivals, accommodation occupancy rates, average length of stay, and visitor spending. Following the recovery from COVID-19, STB management wants to use analytics to improve tourism planning, resource allocation, and marketing targeting. Historical data from 2015–2024 is available, covering variables such as arrival month, visitor nationality, travel purpose, accommodation type, tourism zone (Kuching, Miri, Sibu, rural ecotourism), and spending per category.

Questions:

1. Using descriptive analytics, analyse visitor arrival patterns between 2019 and 2024. Identify key trends, seasonal patterns, and anomalies. Prepare a dashboard summary with at least three visualisations. (25 marks)
2. Apply a diagnostic analytics approach to investigate the key drivers of accommodation occupancy rates. Conduct a correlation analysis and discuss the relationships found. (20 marks)
3. Build a predictive model to forecast visitor arrivals for the next 12 months. Justify your choice of model and interpret the key outputs in non-technical managerial language. (30 marks)
4. Propose a prescriptive analytics solution to help STB optimise marketing budget allocation across visitor segments and tourism zones. Describe the data inputs, model type, and expected managerial outputs. (25 marks)

Report length: 2,500–3,500 words. CLOs assessed: CLO1, CLO2, CLO3.

Sample Assessment 3: Final Assessment — Individual Assignment (30%)

This individual assignment requires you to evaluate the strategic role of analytics in a Malaysian or ASEAN organisation of your choice and to propose a comprehensive analytics strategy.

Task:

Select an organisation from the Malaysian or ASEAN region that has adopted or is in the process of adopting business analytics. Prepare an analytical strategy report (3,000–4,000 words) addressing the following:

- Current analytics maturity assessment: Where does the organisation sit on the analytics maturity curve? (20 marks)
- Critical evaluation of current analytics use: Which analytics types (descriptive/diagnostic/predictive/prescriptive) are being used and how effectively? (25 marks)
- Identification of analytics gaps and opportunities: Where can analytics add the most business value? (20 marks)
- Analytics strategy recommendation: Propose a 3-year analytics roadmap with specific tools, data investments, and capability development plans. (25 marks)
- Ethical and governance considerations: Address data privacy (PDPA), bias in models, and responsible AI considerations. (10 marks)

CLOs assessed: CLO1, CLO2, CLO3.

APPENDIX A

AI and Freemium Analytics Tools for MBA Students

The analytics landscape has been transformed by the emergence of accessible AI-powered tools that allow managers and students to conduct sophisticated analysis without deep coding expertise. The tools below are free or freemium — meaning a full-featured version is available at no cost. They are organised by analytics stage to help you select the right tool for each Learning Unit.

LU1 & LU2 — Data Understanding and Exploratory Analysis

Tool	Access	Cost	How to Use for Analytics
Julius AI	https://julius.ai	Freemium (25 analyses/month free)	Upload any CSV or Excel file and ask questions in plain English. Julius generates charts, summary statistics, and written insights automatically. Ideal for EDA without coding. Example prompt: "Summarise this dataset and identify any outliers in the sales column."
Microsoft Copilot in Excel	https://support.microsoft.com/en-us/copilot-excel	Free with Microsoft 365 (university licence)	Ask Copilot to analyse your Excel data conversationally. Example: "Create a pivot table showing average sales by region and highlight the top 3 performers." Available directly in Excel's ribbon if your institution has Microsoft 365.
Google Gemini in Sheets	https://workspace.google.com/features/gemini/	Free with Google account	Ask Gemini to analyse, summarise, or visualise data directly in Google Sheets. Good for collaborative datasets shared via Google Drive.

Tool	Access	Cost	How to Use for Analytics
Genspark	https://www.genspark.ai	Free	Multi-agent AI for business research and data synthesis. Useful in the Business Understanding phase of CRISP-DM — ask it to research an industry, identify key business metrics, or summarise competitor analytics practices.

LU3 — Diagnostic Analytics and Hypothesis Testing

Tool	Access	Cost	How to Use for Analytics
JASP	https://jasp-stats.org	Fully Free (open-source)	A point-and-click statistical analysis tool — no coding required. Supports t-tests, ANOVA, correlation, regression, and Bayesian analysis. Produces publication-quality output tables. Download and install on any Windows/Mac laptop. Highly recommended for LU3 assignments.
Julius AI	https://julius.ai	Freemium	Upload your dataset and ask: "Run a one-way ANOVA to test whether mean sales differ across the three regions." Julius will run the test and explain the output in plain English.

Tool	Access	Cost	How to Use for Analytics
ChatGPT	https://chat.openai.com	Free (GPT-4o)	Use as an analytics interpreter: paste your ANOVA or t-test output and ask "What does this mean in business terms?" or "Is this result statistically significant at $p < 0.05$?" Highly effective for translating statistical output into managerial language.

LU4 — Predictive Analytics

Tool	Access	Cost	How to Use for Analytics
Orange Data Mining	https://orangedatamining.com	Fully Free (open-source)	A visual, drag-and-drop machine learning platform. Build classification trees, logistic regression models, and clustering pipelines without writing a single line of code. Excellent for learning predictive analytics conceptually. Available on Windows, Mac, and Linux.
Julius AI	https://julius.ai	Freemium	Upload your dataset and ask: "Build a logistic regression model to

			predict customer churn. Show me the confusion matrix and explain the key predictors." Julius handles the modelling and explains results clearly.
Google Teachable Machine	https://teachablemachine.withgoogle.com	Free	A beginner-friendly tool for understanding classification — train a simple image or audio classifier in minutes without any coding. Good for building intuition about how classification models learn.
AutoML by Google (Vertex AI)	https://cloud.google.com/vertex-ai/docs/beginner/beginners-guide	Free tier available	For students wanting to explore production-grade ML without full coding — Google's AutoML trains models on your data automatically. Free credits available for students.

LU5 — Prescriptive Analytics

Tool	Access	Cost	How to Use for Analytics
Excel Solver	Built into Microsoft Excel	Free (part of Excel)	Excel's built-in optimisation add-in. Solves linear programming, integer programming, and non-linear optimisation problems. Enable via: File → Options → Add-ins → Solver Add-in. Directly applicable to the palm oil mill LP example in LU5.
ChatGPT	https://chat.openai.com	Free (GPT-4o)	Use for decision analysis: describe your decision scenario and ask ChatGPT to "Build a payoff table with EMV calculations" or "Help me formulate this as a linear programming problem." Very effective as a prescriptive decision assistant.
Solver Foundation (Excel + Python)	https://www.python.org	Free	For advanced students: Python's PuLP library provides free linear programming capability. Example: pip install pulp then use LpProblem to define and solve LP models programmatically.

LU6 — Data Visualisation and Storytelling

Tool	Access	Cost	How to Use for Analytics
Power BI Desktop	https://powerbi.microsoft.com/en-us/desktop/	Fully Free (download)	Microsoft's industry-standard BI tool. Connect to Excel, CSV, SQL, or cloud data sources. Drag-and-drop dashboard design. Recommended for the dashboard assignment. Large community with free tutorials on YouTube and Microsoft Learn.
Google Looker Studio	https://lookerstudio.google.com	Fully Free	Browser-based dashboard builder. Connects directly to Google Sheets, Google Analytics, and many other data sources. Easy to share via link — no software installation needed. Good for students using Google Workspace.
Tableau Public	https://public.tableau.com	Fully Free	The free version of industry-leading Tableau. Full visualisation capability; dashboards saved to the public Tableau gallery. Note: data is publicly visible, so do not use confidential datasets.

Tool	Access	Cost	How to Use for Analytics
Canva Data Visualisation	https://www.canva.com/graphs/	Free tier available	For students who need to create presentation-quality charts quickly without BI tools. Good for infographic-style data stories and executive presentations. Limited analytical depth but strong visual output.
Julius AI	https://julius.ai	Freemium	Ask Julius to "create a dashboard from this tourism dataset showing arrivals by month and source market." Julius generates Python-powered charts that can be downloaded. Useful for rapid prototyping before building in Power BI.

Responsible Use of AI in Analytics

AI tools are powerful accelerators but must be used responsibly in academic and professional contexts. The following principles apply:

1. Verify all AI-generated outputs: AI tools can make computational errors. Always cross-check results against manual calculations or a second tool before relying on them for decisions or assessments.
2. Declare AI tool use in assignments: i-CATS's academic integrity policy requires you to declare when AI tools were used in assessment preparation. Use the standard declaration form provided by your programme coordinator.
3. AI generates insights, not wisdom: AI tools can identify patterns and calculate statistics, but managerial interpretation and contextual judgement remain human responsibilities. The "so what?" question can only be answered by a manager who understands the business.

4. Data privacy (PDPA): Do not upload confidential, proprietary, or personally identifiable data to cloud-based AI tools (Julius AI, ChatGPT, Gemini, etc.) without verifying the tool's data privacy policy and obtaining appropriate organisational approval.
5. Bias awareness: AI models trained on historical data can perpetuate historical biases. Always critically evaluate model outputs for fairness — particularly in HR analytics, credit scoring, or any application affecting people.

APPENDIX B

Analytics Formula Reference Card

This reference card summarises all key formulas used in MBA6653. Keep this page accessible during practice sessions and self-check exercises.

LU2 — Descriptive Statistics

Measure / Statistic	Formula	Notes
Mean (Average)	$\bar{x} = \sum x_i / n$	Sum all values, divide by count n
Median	Middle value when sorted ascending	For even n: average of two middle values
Mode	Most frequently occurring value	A dataset can be bimodal or multimodal
Range	Range = Max – Min	Simplest measure of spread
Variance (sample)	$s^2 = \sum (x_i - \bar{x})^2 / (n-1)$	Average squared deviation from mean
Standard Deviation	$s = \sqrt{s^2}$	Square root of variance; same units as data
Coeff. of Variation	$CV = (s / \bar{x}) \times 100\%$	Relative spread; enables cross-scale comparison
IQR	$IQR = Q3 - Q1$	Middle 50% of data; robust to outliers
z-score	$z = (x - \mu) / \sigma$	How many standard deviations from the mean

LU3 — Diagnostic / Hypothesis Testing

Measure / Statistic	Formula	Notes
One-sample t-statistic	$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$	df = n - 1
Two-sample t-statistic	$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$	Welch's t-test (unequal variances)
Pearson Correlation	$r = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / [(n-1) \cdot s_x \cdot s_y]$	Range: -1 to +1
ANOVA F-statistic	$F = MS_Between / MS_Within$	MS = SS / df for each source
SS Between (ANOVA)	$SSB = \Sigma n_j (\bar{x}_j - \bar{x})^2$	n_j = group size; $\bar{x}_j$ = group mean
SS Within (ANOVA)	$SSW = \Sigma \Sigma (x_{ij} - \bar{x}_j)^2$	Sum of squared deviations within each group

LU4 — Predictive Analytics

Measure / Statistic	Formula	Notes
Simple Linear Regression	$\hat{Y} = \beta_0 + \beta_1 X$	β_0 = intercept; β_1 = slope
Slope (β_1)	$\beta_1 = r \times (s_y / s_x)$	r = correlation; s = std deviation
Intercept (β_0)	$\beta_0 = \bar{y} - \beta_1 \bar{x}$	Calculated after finding β_1
R-squared	$R^2 = 1 - (SSE / SST)$	Proportion of variance explained by model
SSE (Residual)	$SSE = \Sigma (y_i - \hat{y}_i)^2$	Sum of squared prediction errors

Measure / Statistic	Formula	Notes
RMSE	$RMSE = \sqrt{(SSE / n)}$	Root Mean Squared Error — prediction accuracy
Accuracy (Classification)	$Accuracy = (TP + TN) / Total$	Correct predictions as % of all predictions
Precision	$Precision = TP / (TP + FP)$	Of predicted positives, how many are correct?
Recall (Sensitivity)	$Recall = TP / (TP + FN)$	Of actual positives, how many were found?
F1-Score	$F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$	Harmonic mean of Precision and Recall
Association: Support	$Support(A \cap B) = A \cap B / N$	Frequency of itemset in all transactions
Association: Confidence	$Confidence(A \rightarrow B) = Support(A \cap B) / Support(A)$	Probability B given A
Association: Lift	$Lift(A \rightarrow B) = Confidence(A \rightarrow B) / Support(B)$	Lift > 1 = positive association

LU5 — Prescriptive Analytics

Measure / Statistic	Formula	Notes
LP Objective Function	Maximise/Minimise $Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$	c_i = objective coefficient; x_i = decision variable
LP Constraints	$a_{i1}x_1 + a_{i2}x_2 + \dots \leq b_i$ (for all i)	Each constraint limits available resources
Expected Monetary Value	$EMV = \sum [P(s_j) \times Payoff(d_i, s_j)]$	$P(s_j)$ = probability of state j; sum over all states

Measure / Statistic	Formula	Notes
EOL (Expected Opp. Loss)	$EOL = \sum [P(s_j) \times \text{Regret}(d_i, s_j)]$	Minimise EOL $\equiv$ Maximise EMV
Monte Carlo Output	Distribution of NPV / outcome metric	Run 1,000+ iterations; report mean, SD, percentiles

Key Statistical Notation

Symbol	Meaning
$\bar{x}$ (x-bar)	Sample mean
μ (mu)	Population mean
s	Sample standard deviation
σ (sigma)	Population standard deviation
s^2 or σ^2	Variance
n	Sample size
N	Population size
H_0	Null hypothesis
H_1 or H_a	Alternative hypothesis
α (alpha)	Significance level (typically 0.05)
p-value	Probability of observed result given H_0 is true
df	Degrees of freedom
r	Pearson correlation coefficient
R^2	Coefficient of determination
TP/TN/FP/FN	True Positive/Negative, False Positive/Negative

APPENDIX C

Optional Advanced Track: Python and R Code Snippets

For students who wish to go beyond Excel and no-code tools

This appendix is entirely optional. It provides Python and R code snippets that replicate the analyses taught in each LU. No prior programming experience is required to complete the MBA6653 assessments — all analyses can be performed using Excel, JASP, Julius AI, or Power BI. However, for students who wish to develop programming capability, these snippets provide a direct bridge between course concepts and industry-standard analytics languages.

Getting Started:

1. Python: Install Anaconda (<https://www.anaconda.com>) — a free distribution that includes Python, Jupyter Notebook, and all required libraries (pandas, numpy, scipy, sklearn, matplotlib, seaborn). No separate installation of packages needed.
2. R: Install R (<https://www.r-project.org>) and RStudio Desktop (<https://posit.co/download/rstudio-desktop/>) — both free. RStudio provides a user-friendly interface for writing and running R code.
3. Google Colab (<https://colab.research.google.com>): Free cloud-based Python notebook — no installation required. Run Python code directly in your browser. Recommended for students without a laptop that can run Anaconda.

LU2 — Descriptive Statistics in Python (Python)

Calculate all key descriptive statistics for a dataset using pandas.

```
import pandas as pd
import numpy as np

# Load your data (replace with your actual file path)
df = pd.read_csv("delivery_times.csv")

# Summary statistics
print(df["delivery_hours"].describe())

# Additional measures
print("Mean:", df["delivery_hours"].mean())
print("Median:", df["delivery_hours"].median())
print("Mode:", df["delivery_hours"].mode()[0])
print("Std Dev:", df["delivery_hours"].std())
print("CV:", df["delivery_hours"].std() / df["delivery_hours"].mean() * 100,
      "%")
print("IQR:", df["delivery_hours"].quantile(0.75) -
      df["delivery_hours"].quantile(0.25))

# Histogram
import matplotlib.pyplot as plt
df["delivery_hours"].hist(bins=10, color="#1A4E8C", edgecolor="white")
plt.title("Distribution of Delivery Times")
plt.xlabel("Hours"); plt.ylabel("Frequency")
plt.tight_layout(); plt.show()
```

LU2 — Descriptive Statistics in R (R)

Equivalent R code using base R and ggplot2.

```
# Load data
df <- read.csv("delivery_times.csv")

# Summary statistics
summary(df$delivery_hours)
sd(df$delivery_hours)

# CV and IQR
cv <- sd(df$delivery_hours) / mean(df$delivery_hours) * 100
iqr_val <- IQR(df$delivery_hours)
cat("CV:", cv, "%\nIQR:", iqr_val, "\n")

# Histogram with ggplot2
library(ggplot2)
ggplot(df, aes(x=delivery_hours)) +
  geom_histogram(bins=10, fill="#1A4E8C", color="white") +
  labs(title="Distribution of Delivery Times",
       x="Hours", y="Frequency") +
  theme_minimal()
```

LU3 — Hypothesis Testing (t-test and ANOVA) in Python (Python)

Run a one-sample t-test and one-way ANOVA using `scipy.stats`.

```
from scipy import stats
import pandas as pd

# One-sample t-test
# H0: mean moisture content >= 15%; H1: mean < 15%
sample = [13.2, 14.1, 12.8, 13.9, 14.5, 13.1, 12.9, 14.2, 13.7, 13.5,
          14.0, 13.3, 12.7, 14.3, 13.8, 13.6, 14.1, 12.9, 13.4, 14.0,
          13.2, 13.7, 12.8, 14.4, 13.9]

t_stat, p_value = stats.ttest_1samp(sample, popmean=15, alternative="less")
print(f"t-statistic: {t_stat:.3f}")
print(f"p-value (one-tail): {p_value:.4f}")
print("Reject H0" if p_value < 0.05 else "Fail to Reject H0")

# One-way ANOVA — comparing sales across 3 cities
kuching = [45, 52, 48, 55, 49, 51, 47, 53]
miri     = [38, 42, 40, 44, 39, 43, 41, 45]
kk       = [41, 46, 43, 48, 42, 47, 44, 50]

f_stat, p_anova = stats.f_oneway(kuching, miri, kk)
print(f"\nANOVA F-statistic: {f_stat:.3f}")
print(f"p-value: {p_anova:.4f}")
print("At least one group differs" if p_anova < 0.05 else "No significant difference")
```

LU4 — Linear Regression in Python (Python)

Build and interpret a simple linear regression model using sklearn.

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
import matplotlib.pyplot as plt

# Sample data: daily max temperature vs peak demand (MW)
temperature = [28, 31, 33, 30, 35, 29, 32, 34, 36, 31, 33, 30,
               34, 37, 32, 29, 31, 35, 33, 36, 30, 34, 28, 32, 35]
peak_demand = [967, 1020, 1065, 1009, 1095, 985, 1046, 1078, 1120,
               1017, 1056, 1004, 1082, 1138, 1043, 988, 1023, 1098,
               1061, 1115, 1001, 1079, 955, 1048, 1103]

X = np.array(temperature).reshape(-1, 1)
y = np.array(peak_demand)

model = LinearRegression().fit(X, y)

print(f"Intercept ( $\beta_0$ ): {model.intercept_:.2f}")
print(f"Slope ( $\beta_1$ ): {model.coef_[0]:.2f} MW per °C")
print(f"R²: {r2_score(y, model.predict(X)):.4f}")
print(f"RMSE: {mean_squared_error(y, model.predict(X), squared=False):.2f} MW")

# Predict peak demand for 38°C day
forecast = model.predict([[38]])
print(f"\nForecast for 38°C: {forecast[0]:.0f} MW")
```

LU4 — K-Means Clustering in Python (Python)

Segment SMEs into clusters by digital adoption, revenue growth, and employee count.

```
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt

# Load SME dataset (columns: digital_score, revenue_growth_pct, employees)
df = pd.read_csv("sarawak_sme_data.csv")

# Standardise features (important for K-Means)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df[["digital_score", "revenue_growth_pct",
"employees"]])

# Elbow method to find optimal k
wcss = []
for k in range(1, 9):
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_scaled)
    wcss.append(km.inertia_)

plt.plot(range(1, 9), wcss, marker="o")
plt.title("Elbow Method – Optimal k")
plt.xlabel("Number of Clusters"); plt.ylabel("WCSS")
plt.tight_layout(); plt.show()

# Fit final model with k=3
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
df["cluster"] = kmeans.fit_predict(X_scaled)

# Profile each cluster
print(df.groupby("cluster")[["digital_score", "revenue_growth_pct", "employees"]].mean())
```

LU5 — Linear Programming in Python (PuLP) (Python)

Solve the palm oil mill LP optimisation problem from LU5.

```
# Install first: pip install pulp
from pulp import *

# Define the problem
prob = LpProblem("PalmOilMill", LpMaximize)

# Decision variables
cpo = LpVariable("CPO_tonnes", lowBound=0)
pko = LpVariable("PKO_tonnes", lowBound=0)

# Objective function: Maximise profit
prob += 400 * cpo + 300 * pko, "Total Profit"

# Constraints
prob += 3 * cpo + 2 * pko <= 180, "Processing Hours"
prob += 2 * cpo + 3 * pko <= 240, "FFB Supply"

# Solve
prob.solve(PULP_CBC_CMD(msg=0))

print(f"Status: {LpStatus[prob.status]}")
print(f"Optimal CPO: {cpo.varValue:.1f} tonnes")
print(f"Optimal PKO: {pko.varValue:.1f} tonnes")
print(f"Maximum Profit: RM {value(prob.objective):.0f}")
```

LU6 — Data Visualisation in Python (Python)

Create a multi-panel dashboard-style figure using matplotlib and seaborn.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Sample tourism data
months =
["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]
arrivals_2024 = [420, 487, 510, 445, 390, 360, 480, 520, 505, 475, 430, 610]
arrivals_2023 = [356, 413, 432, 378, 331, 306, 407, 441, 428, 403, 365, 518]

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Chart 1: Arrivals trend – 2023 vs 2024
axes[0].plot(months, arrivals_2024, marker="o", color="#1A4E8C", label="2024",
linewidth=2)
axes[0].plot(months, arrivals_2023, marker="s", color="#B8860B", linestyle="--",
label="2023")
axes[0].set_title("Sarawak Visitor Arrivals Up 18% YoY", fontsize=12,
fontweight="bold")
axes[0].set_ylabel("Arrivals (thousands)"); axes[0].legend();
axes[0].grid(axis="y", alpha=0.3)

# Chart 2: Top source markets – horizontal bar
markets = ["Singapore", "Brunei", "Indonesia", "Australia", "China", "UK"]
counts = [185, 92, 74, 61, 48, 35]
colours = ["#1A4E8C", "#1A4E8C", "#1A4E8C", "#AAAAAA", "#AAAAAA", "#AAAAAA"]
axes[1].barh(markets, counts, color=colours)
axes[1].set_title("Top Source Markets – YTD 2024", fontsize=12,
fontweight="bold")
axes[1].set_xlabel("Arrivals (thousands)")
for i, v in enumerate(counts): axes[1].text(v+1, i, str(v), va="center")

plt.tight_layout()
plt.savefig("sarawak_tourism_dashboard.png", dpi=150, bbox_inches="tight")
plt.show()
```


Further Learning Resources for Python and R

1. Python for Data Analysis (Wes McKinney) — The definitive pandas guide. Available free online via O'Reilly if your institution has a subscription.
2. Kaggle Learn (<https://www.kaggle.com/learn>) — Free, interactive Python courses on Pandas, Data Visualisation, ML. Complete in-browser with no installation.
3. DataCamp (<https://www.datacamp.com>) — Structured Python and R courses for data analytics. Free tier available; full access often included in university subscriptions.
4. R for Data Science (Hadley Wickham) — Available free at <https://r4ds.hadley.nz>. The best introduction to data analysis in R using the tidyverse.
5. Google Colab (<https://colab.research.google.com>) — Free cloud Jupyter notebooks with GPU access. All code in this appendix can be run directly in Google Colab.